- **Confidence Level** : It is the confidence with which a person is sure that the True value lies in b/w the given confidence interval.

- Here the value of Standard Deviation (s) is NOT known. & it is evaluated from the sample itself.

$$s = \sqrt{\frac{\Sigma (x - \bar{x})^2}{n-1}}$$

Here x = Individual Values of Obs. Term.

∴ Value of n is very small, we refer to the "Student t - distribution rather than the normal distribution".

Due to this transition to the t - distribution table, the NEW CONFIDENCE INTERVAL is:

$$\text{Te lies in } \left(\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}} \right) = (1-\alpha)$$

Then WKT-

$$K\bar{x} = t_{\alpha/2} \frac{s}{\sqrt{n}}$$

$$\therefore \quad n = \left[\frac{t_{\alpha/2}\ s}{K\bar{x}}\right]^2$$

fill here for derivation

where n = no. of cycles

$t_{\alpha/2}$ = value found from 't' distribution

$\bar{x}$ = mean value

s = Standard value

K = Internal size (eg ± 10%)

Q. How to read the value of $t_{\alpha/2}$?

Ans ① Note down the already no. of performed cycles as 'N'.

② Degree of freedom, dof = N − 1.

③ Then the value of 2α will be given as the confidence interval.

④ Calculate the value for $\alpha/2$.

⑤ Check the req values against α & dof column.

Q. Can the analyst determine the size of the confidence interval at certain confidence levels?

Ans: Yes. the formula used will be -

$$K = \frac{t_{\alpha/2}\, s}{\bar{x}\sqrt{n}}$$

Note: More the no. of Obs. more the accuracy.

⟹ Performance Rating -

- aka "Performance Leveling"
- Simplest & most common method is based on Speed / Pace.

Q. What is Speed Rating?

Ans: It is the Rating of the worker's pace against the standard performance.

- Pace is the ONLY factor that is Rated.
- The concept of Standard Performance is different for different kinds of work.

↳ Aspects to be considered during Rating are -

① Degree of difficulty for that task.

② Worker's Pace relative to the concept of 100% Pace.

↳ Training of an Pace Recorder ?

- Training is done by the use of Videotapes. of workers performing a task at various performance levels.

- Based on the type & length of work, the analyst then records the rates of performances as: ① Elemental Ratings

(or)

② Overall Ratings.

Q. What are the characteristics of a well implemented performance rating system ?

Ans: ① Consistency among Tasks.

② Consistency among Analysts.

(3) Easily understood.

(4) Related to Standard Performance.

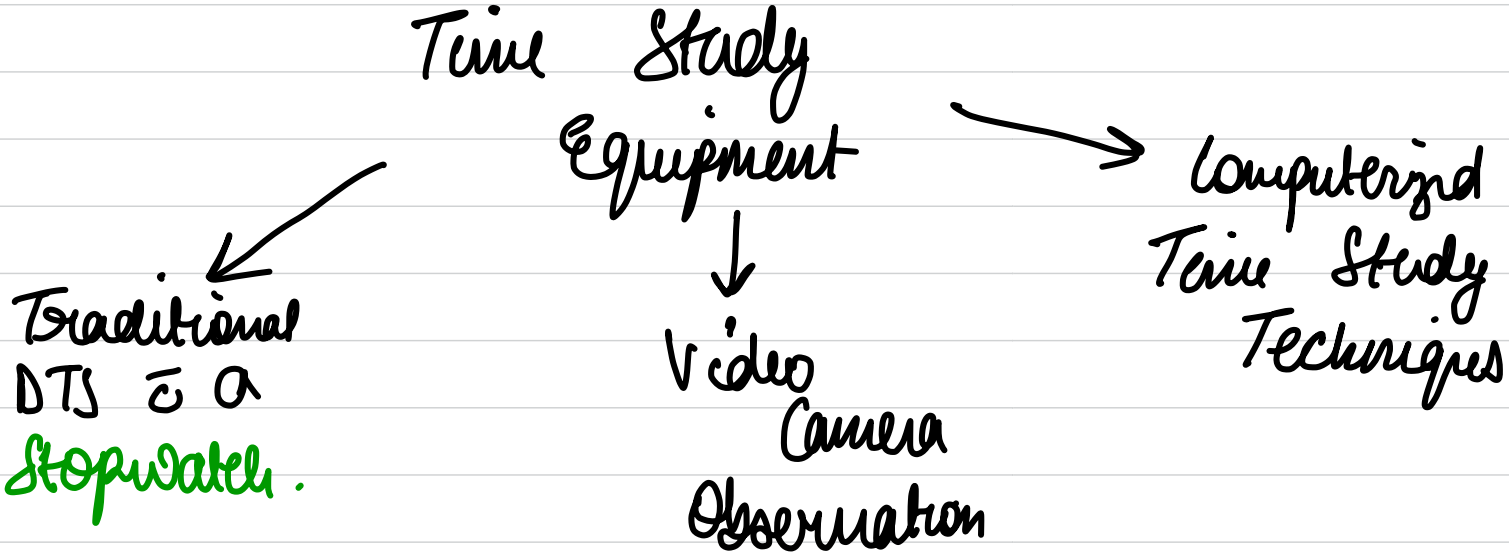
(5) Machine paced elements rated @ 100%

(6) Performance rating recorded during obs. of task.

(7) Work Notification.

⟹ Time Study Equipment:

It is classified into 3 techniques ie -



(1) Traditional DTS with a Stopwatch:-

- It consists of a Stopwatch & Time Study form.

Stopwatches can be Classified as -

(i) Mechanical Stopwatches

(ii) Electronic Stopwatches.

↳ ① Mechanical Stopwatches -

- They are typically graduated in 3 time scales:

① Decimal Minutes (ie 0.01 min)

② Decimal Hour

③ TMU (Time Measurement Units)

ie [1 TMU = 0.00001 hr = 0.036 sec]

- Different scales are also used, since
  - Stopwatch of Decimal Minute are very common.
  - Stopwatch of Decimal Hour are LESS common
  - Stopwatch of TMU so it complies with an organization's PMTS.

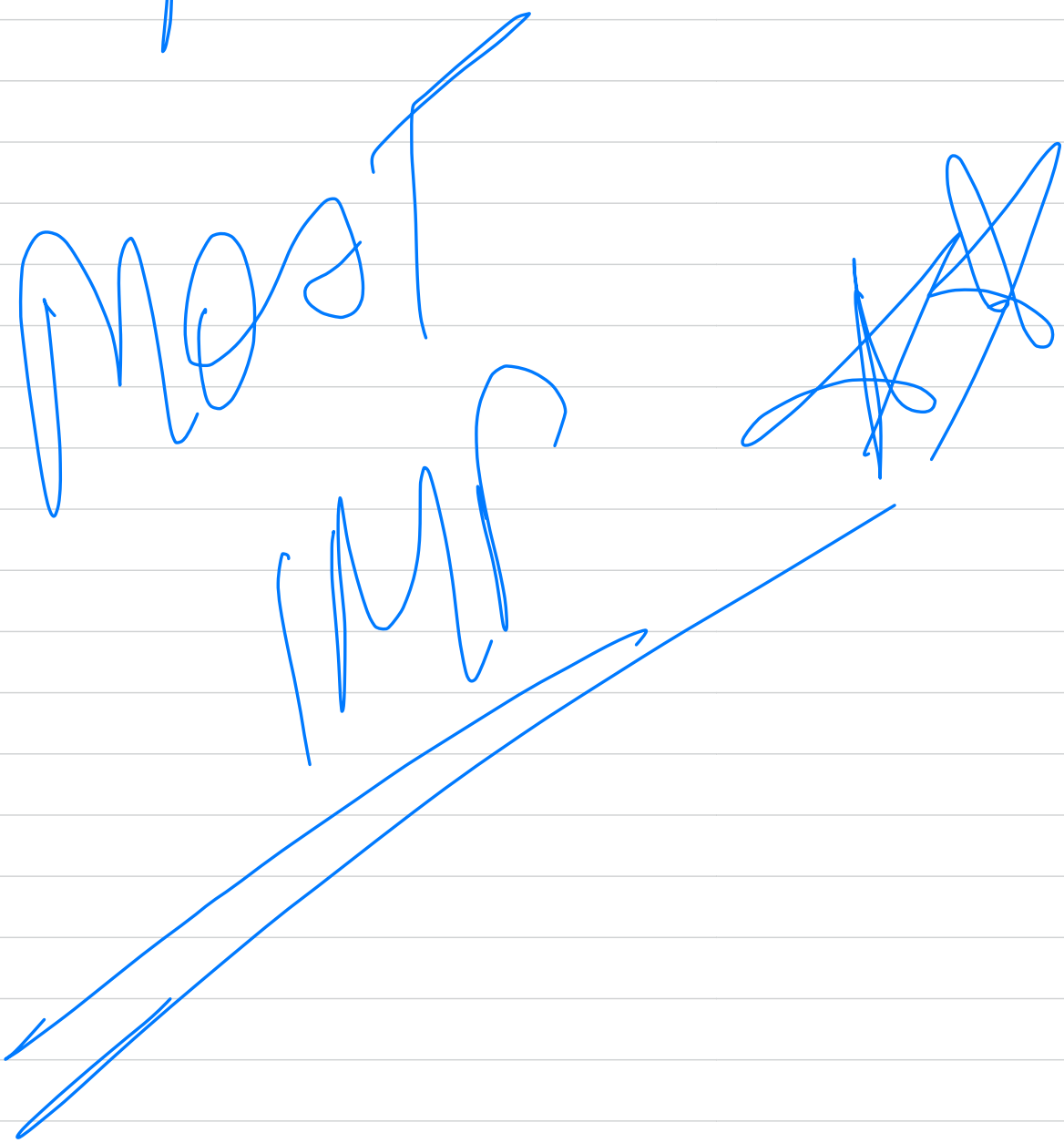
↳② Electronic Stopwatch.

- It provides a Digital display
- It is easier to read the time.
- There are less reading errors.
- They are light in weight.
- They are more ACCURATE & PRECISE.

↳② Video camera Rating -

- It provides for a A&V record of the method used by the worker.
- The video also provides an accurate record of times taken by each work element.
- The tape can be used to settle disputes about the standard method, performance & standard time.
- Tapes can be used to Train New Time Study analysts.

Solve Numerically



(III) Predetermined Motion Time Systems (PMTS).

- Methods Study & Work Measurement are complementary to each other.
- PMTS is preffred for the Levels of "BME & Work elements".

Def: A PMTS is a DATABASE of BME's & their associated normal time values.

- Here the Normal Times are determined based on Extensive Research.
- Frank Gilberth Suggested Manual work cycles of BME.
- Segur developed the 1st ever PMTS i/s MTA by analysing motion picture films.
- Joseph Quick developed 1st Work-Factor System.
- MTM (ie 1st Level PMTS) was developed by Maynard.
- Higher Level PMTS were built from MTM.

- Bailey & Presgrave developed "Basic Motion Timestudy" (BMT)
- Gressan & Nance developed "MASTER Standard Data" (MSD)
- Heyde developed "Modular Arrangement of Predetermined Time Standards" (MOD APT).
- Zadin developed MOST.

↳ Procedure of PMTS?

Q. what is the Procedure to Establish a PMTS?

Ans: (1) Synthesize the Methods.

(2) Retrieves all the Normal Times for each BME.

(3) Evaluate the method improvement (Obj of the Evaluation is to reduce NT)

(4) Apply allowances & prepare a DATABASE with Standard Times of each Element.

Note - Usual PMTS do not include "ALLOWANCES" & allowance must be incorporated by the management.

Note: Here PR is not required ∵ The BME are already rated to 100% performance.

- MTM & MOST use the defn of "DAYWORK" standard performance.
- PMTS other than MTM & MOST, require an adjustment for correct data values.

Eg - MTA is based on "Faster Working Pace" & is faster by about 20% than MTM.

→ Levels & Generations of PMTS:

It is of mainly 2 types-

(i) 1st Level PMTS - Use BME

Eg - Transport Loaded, Use, Position etc.

- They are very detailed.

Eg - MTM - 1 is a 1st Level PMTS.

- Normal Times Values are very small for each BME.
- 1st Level PMTS employ High Application Speed Ratio.
- 1st Level PMTS are MOST ACCURATE.

(ii) Higher Level PMTS - They make use of "Aggregates"

Q. What is an Aggregate?

Ans: Aggregate is a combn of 2 or more BME.

Eg GET = Reach + Grasp

Get is a Work Element

BME's.

- Higher Level PMTS are "CONDENSED DATA-BASES" with fewer body motions.

Eg - MTM-2 : 2nd Generation MTM.

- For Higher Level PMTS, the Application speed ratio is small.
- Higher Level PMTS are not as ACCURATE as 1st Level PMTS.
- MOST relates to Higher Level PMTS.

PTO

Q. What are the Characteristics of PMTS levels?

Ans: Parameters of Comparison are -

Characteristics	1st Level	Higher Level
(1) Accuracy.	Most Accurate	Least Accurate.
(2) Application Speed Ratio	High	Less
(3) Suitability on the Type of Task	Repetative	Non-Repetitive
(4) Cycle Time	Shorter ie (less than 1 min)	Longer (ie More than 1 min)
(5) Type of Elements	BME	Aggerates
(6) Method's description	Very Detailed	Less Detailed
(7) Flexibility of Application	Very Flexible	Less Flexible

→ Types of PMTS.

Q. Elucidate on the different types of PMTS?

Ans: They are 7 types of PMTS -

1. Motion Time Analysis (MTA) -
2. Work Factor (WF)
3. Methods Time Measurement (MTM) -
4. Basic Motion Time Study (BMT)
5. Master Standard Data (MSD)
6. Maynard Operations Sequencing Technique (MOST)
7. MODular Arrangement of Predetermined Time Standards (MODAPTS)

⟹ Methods Time Measurements (MTM) -

Def: "MTM is a procedure which analyses any manual operation into basic motions & assigns a predetermined standard to each motion."

→ What are the Types of MTM?

↳ MTM Family -

① MTM - 1 :

- It operates at the basic motion elements level.
- Many of the BME's correspond to "Therbligs".
- Unit of Measurement of MTM = 1 TMU (Time Measurement Unit).

∴ 1 TMU = 0.036 seconds

- Symbols used in MTM-1 are -

① Reach (R) (aka) => Transport Empty (TE)

② Grasp (G)

③ Move (M) (aka) => Transport Loaded (TL)

④ Position (P)

⑤ Release (RL)

⑥ Disengage (D)

⑦ Turn (T)

(8) Apply Pressure (AP)

(9) Eye Travel (ET)

(10) Eye Focus (EF)

(11) Simultaneous Motions.

(2) MTM 2 : • It consists of MOTION Aggregates.

• Most Imp. Categories of MTM-2 are : (1) Get = Reach + Grasp

(2) Put = Move + Position.

(3) MTM - 3 : • It has only 4 main Categories -

(i) Handle

(ii) Transport

(iii) Step & Foot Motion

(iv) Bend & Arise.

(4) MTM - UAS : It is used in "BATCH Production"

(5) MTM - MEK : It is used in "Small Lot production with long cycle Times"

(6) MTM - HC : It is used in the Health Care sector.

(7) MTM-C : It is used mostly in "CLERICAL work".
It is of 2 types -

(1) MTM-C_1 : Emphasis on "Precision"

(2) MTM-C_2 : Emphasis on "Application Speed Ratio"

(8) MTM-V : It is used for WM in "Machine Tool operations"

(9) MTM-TE : It is used for WM in "Electronic Testing Applications"

(10) MTM-M : It is used for measuring assembly work under "Stereoscopic Microscope"

⇒ Maynard's Operation Sequence Technique (MOST)

- It is a High Level PMTS based on MTM.
- It uses the same unit as that of PMTS i.e TMU

- MOST focus on the use of "Aggregates".

Q. What is the Essence of MOST (ie Primary Obj):

Ans: FOCUS: "Analysis of Work Activities mirroring movement of "OBJECTS."

If there are NO PHYSICAL OBJECTS to move then the moments are considered as "BODY MOVEMENTS".

- These MOVEMENTS are tracked by the use of "AGGREGATES"
- Aggregates aka => ACTIVITY Sequence MODELS.

⇒ Varities of MOST -

- It is Classified into 4 main Varities:

1. BASIC MOST
2. MINI MOST
3. MAXI MOST
4. Clerical MOST.

(I) Basic Most -

- Here the movement of Objects consists of Pattern of body motions.
- It uses "MOTION AGGREGATES" ie also k/s "ACTIVITY SEQUENCE MODEL".
- Rarely would a work Element be defined as less than a COMPLETE ACTIVITY sequence model.

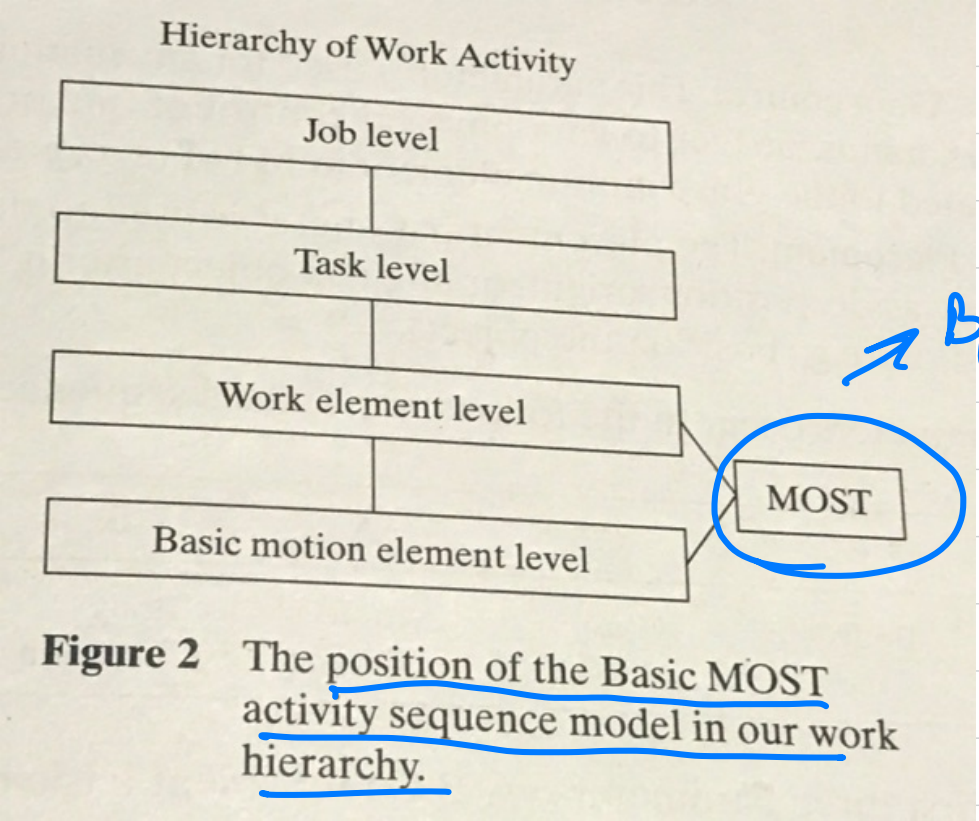


Figure 2 The position of the Basic MOST activity sequence model in our work hierarchy.

↗ B/w WE Level & BME Level.

↳ Categories of Activity Sequence models for Basic - MOST is -

(1) General Move (2) Controlled Move

(3) Tool Use.

- The Basic ACTIONS in Activity Sequence Modelling is also "Sequence Model Parameter".

(i) General Move :

- This Sequence Model is used when an Obj is : "Moved Freely through Space from 1 location to another location"

Eg: Placing Materials, Removing c/ps etc.

- There are 4 sequence model parameters ie (ACTIONS) that are -

(1) A = Action Distance -

- Describes Movement of Finger, Feet, Hands etc.
- Movement can be performed "Loaded or Unloaded"
- It is usually Horizontal.

(2) B = Body Motion - Defines Vertical Body Motions Eg: Sitting, Standing

(3) G = Gain Control

- It is used to describe the manual action of fingers, feets & hands to gain PHYSICAL control over an object.
- Gain Control is closely related to Grasp of MTM.

(4) P = Placement -

- Describes the actions of aling position orient. lay aside of the object @ a "NEW LOCATION".

∴ Standard Sequences of General move one -

ie Get, Put, Return

when Get = A B G

Put = A B P

Return = A

∴ Standard Sequences of Genral move is -

A B G A B P A

very IMP ***

(ii) Controlled Move -

Def: "The object is moved when it is in contact with the surface".

Eg - Sliding

- Controlled Move Sequence models are used when an OBJECT is moved through a PATH that is constrained -
- It has 6 Sequence Parameters -

(1) A = Action Distance

(2) B = Body Motion

(3) G = Gain Control / Grasp.

(4) M = Move = Describes any motion required to move an Object on a Controlled Path.

(5) X = Process Time = No manual motions are included.

(6) I = Align. = It is used for manual motions done @ the End of the Controlled Move Sequence.

∴ Standard sequence of controlled moves is -

Get , Actuate & Align , Return

where Get = A B G

Actuate & Align = M X I

Return = A

∴ Standard sequence in controlled moves is -

A B G M X I A VVVV IMP

(iii) Tool Use -

Def = This sequence model is applied to the use of "HAND Tools."

Hand Tools covered are -
- Screwdrivers
- Hammers
- Scissors
- Knifes
- Wrenches
- Common Measuring instruments.

- It uses 11 sequence Parameter factors -

1. A = Action Distance
2. B = Body Motion
3. G = Gain Control
4. P = Placement
5. F = Fasten = For Fasting Motion Eg - Screwdrivers
6. L = Loosen = For unfasting motion Eg - Wrenches
7. C = Cut = Slice, Slit Eg - Knifes.
8. S = Surface Treatment = To apply a coat on the surface Eg: Brush.
9. M = Measure : Eg - Scales, Gauges
10. R = Record : Use of Pen, Pencils to take readings
11. T = Think : Visual & Mental Activities to Read & Inspect an Obj.

(V) * — (*) can take any of these fns

∴ Standard Sequence for Tool Use is -

Get, Put, Tool (*), Put, Return

where Get = ABG

Put = ABP

Tool (*) : F, L, C, S, M, R, T

Return = A.

∴ Standard Sequence for Tool Use is -

ABG ABP * ABPA

VVVIMP !!!.

(II) MINI MOST:

- It is applicable when the WORK CYCLES are highly "REPEATATIVE" & are performed more than 1500 times/week.
- It uses 2 sequence Modes that are -
  1. General Move
  2. Controlled Move.

(III) MAXI MOST:

- It is used when the "WORKCYCLE" is done at a performance of "LESS than 100 times / week"

- Here there are observable variations in the W/P.
- It consist of 5 Sequence Models -
  1. Part Handeling
  2. Tool Use
  3. Machine Handeling
  4. Transported with Power Equipment
  5. Transported with Wheeled Trucks.

(IV) CLERICAL MOST:

- Clerical Most is the SAME as "BASIC MOST"
- But it is used in terms of "Clerical Operations"

⇒ MOST for Windows OS (computerized MOST).

- MOST for windows can ONLY use 3 of the 4 varities of MOST:
  - ie • BASIC MOST
  - MINI MOST
  - MAXI MOST.
- It is comprised of 2 modules that are -

(i) QUICK MOST

(ii) DIRECT MOST.

↳ (1) Quick Most : (ie Semi Automated Version)

- It immediately provides the - i) Method description
  - ii) Activity Sequence Model
  - iii) Associated Normal Time

→ (2) Direct MOST - (ie Automated Version)

- It is feature, that AUTOMATICALLY generates the -
  i) Method's Description
  ii) Activity sequence model.
- It allows the analyst to OVERRIDE / EDIT the automatically generated "Method's Description."

Q. What are BENEFITS of MOST for windows?

Ans: Benefits include:

(1) Complete Method's Description

(2) Identification of Non-Productive Elements

(3) Simple to use & Easy to work.

(4) Development of Time Standards is - "Efficient & Accurate".

UNIT-4

(I) Standard Data Systems (SDS) -

(II) Work Sampling -

(III) Learning Curves -

(I) Standard Data Systems (SDS)

Def: It is a DATABASE of NORMAL TIME values usually organised by WORK ELEMENTS.

- Normal Time Values are usually compiled from PREVIOUS DT Studies.
- It is NOT REQUIRED to directly observe the task.
- Standards can be determined BEFORE the job is running.
- If HIGHLY ACCURATE DATA is req then it is source from previous DTs & PMTs.

Eg - If the system is used with an Wage INCENTIVE SYSTEM, then ENGG stds are required.

- If LOW ACCURACY DATA is req, then it is sourced from previous WS, Historical Records & Estimations.

Q. Purposes of using a PMTS to develop a SDS?

Ans: The purpose is -

(1) To SUPPLEMENT the database of DTS values, as some data might be missing.

(2) To DEVELOP the SDS database from the BME times in PMTS.

Note: "The (2) Purpose is DISTINGUISHED from the conventional application of PMTS"

Q. What are MACROSCOPIC & MICROSCOPIC elements?

Sol- With respect to the WORK HIERARCHY.

(1) SDS employes WORK ELEMENTS & hence the ELEMENTS of SDS/Work Elements level are A/S "MACROSCOPIC ELEMENTS"

(2) PMTS employes BME & hence the elements of PMTS/BME level are A/S "MICROSCOPIC Elements"

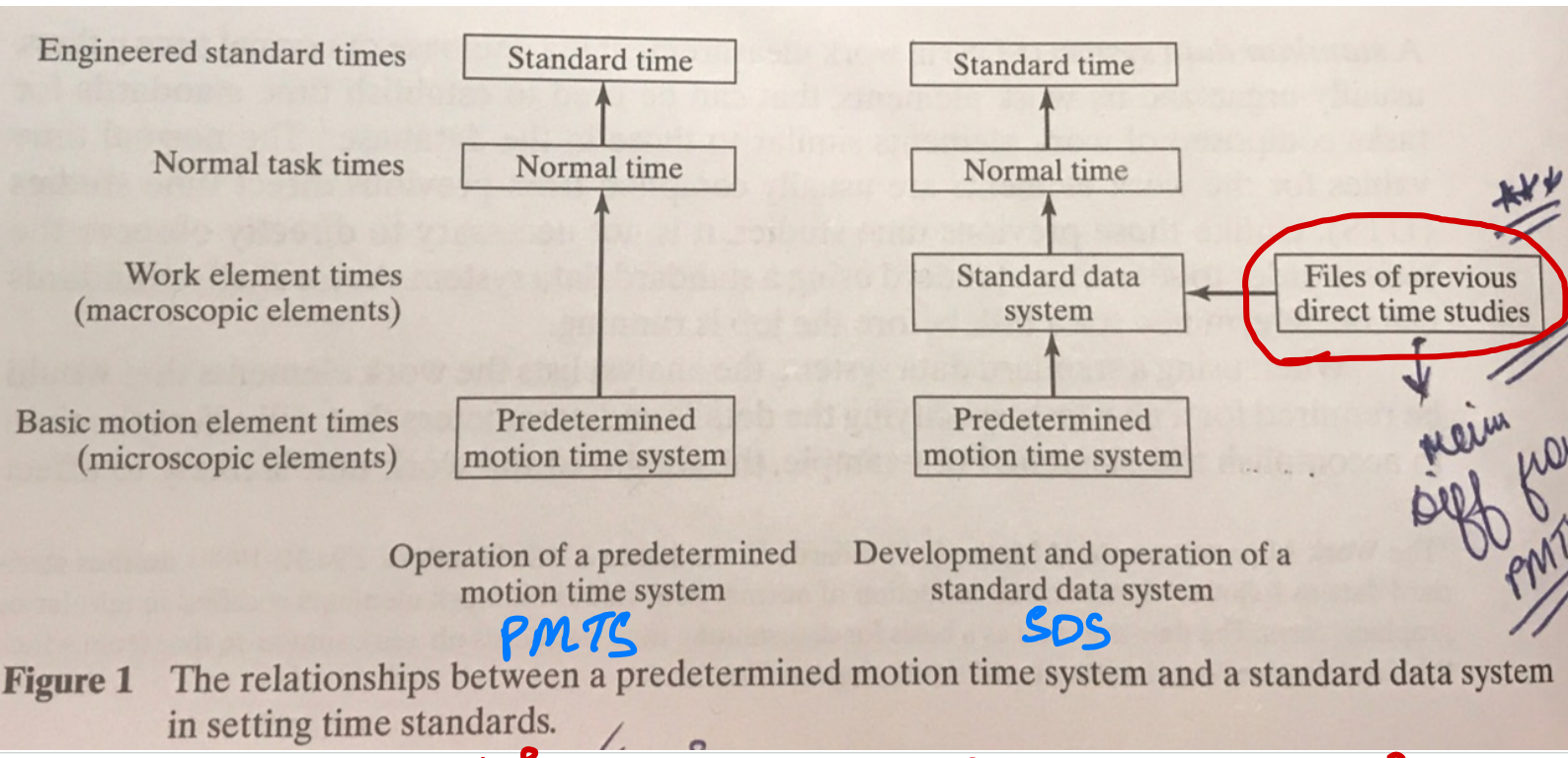


Figure 1 The relationships between a predetermined motion time system and a standard data system in setting time standards.

↳ The main difference b/w PMTS & SDS is -

- PMTS is made with reference to only 1 data-base.
- SDS is made with reference to PMTS & DTS databases.

Q. General Characteristics & Requirements of work situation.

Q. Conditions that warrant the use of SDS.

sol- SDS is most preffered for the following scenarios :

1. SIMILARITY in TASKS
2. BATCH PRODUCTION

(3) LARGE NO. OF STANDARDS to be SET.

(4) NEED TO SET STANDARDS BEFORE PRODUCTION.

- SDS is usually suited to MEDIUM PROD^{n} of BATCHES.
- Note - "Very SHORT CYCLES & HIGHLY Rep. Ops are not suited for SDS."
- When the NO. OF STANDARDS to be set is GREATER than analyst available, then SDS must be opted for.

→ IMPORTANCE of Coverage -

- OBJECTIVE of SDS - "Set Time Standards".
- COVERAGE = It refers to the FAMILY of TASKS for which the SDS is applicable.

Note - • SDS is NOT a universal system.

• SDS is designed for LIMITED RANGE of TASKS.

∴ The coverage must be clearly STATED in the SDS documentation

Q. How to determine Time Standards using SDS. ?

sol- The analyst follows the following steps -

(1) ANALYSE the new task & DIVIDE into WE.

(2) ACCESS the database to determine the req. values of NORMAL Time of WE.

(3) ADD ELEMENT NT to OBTAIN TASK NT
- Setup & Production tasks are computed separately.
- Setup Normal Time = $T_{n\text{-}su}$.
- Prodn Normal Time = T_n.

(4) COMPUTE Standard Times for SETUP & PRODUCTION cycles.

- Standard Time for SETUP is -

$$T_{std\text{-}su} = T_{n\text{-}su}\,(1+Apfd)$$

- Standard Time for PRODUCTION is -

(1) $T_{std} = T_n\,(1+Apfd)$

(2) $T_{std} = T_{nw}(1+Apfd) + T_m(1+Am)$ → For Mk elements.

⇒ SDS DATABASE:

- It contains the NT values of WE that are achieved in a facility.
- It is a CATALOUGE of NORMAL TIME values.
- Work Elements are performed under various CONDITIONS & PARAMETRES.
- These CONDITIONS & PARAMETRES are v/s "WORK VARIBLES".
- WORK VARIBLES relate to the unit work characteristics, TASK parameters & working conditions that influence NT values.
- Different VARIETIES of DATABASE formats can be used.
- Most common formats include -

1. Charts
2. Tables
3. Mathematical formula.
4. Worksheets
5. Computerized DATABASES
6. RETRIVAL Systems.

↳① Charts - It is a Graphical Plot & it indicates -

① NT value : DEPENDENT VARIBLE.

② Work Variables : INDEPENDENT VARIBLE.

It indicats NT values as a fn of the work variables.

↳② Tables - It lists the NT values in an Tabular Arrangement.

- Howeu the PROBLEM ∈ Tables is - It provides the NT value only for DISCRETE values of WV.

↳③ Mathematical Formula - They are a more CONSISE way of expressing the same relation that are presented in Charts & Tables.

↳④ Worksheet - It is a FORM which indicats the Steps the analyst must follow to compute the standards.

↳⑤ Computerized Databases & Retrival Systems - It is a MODERN fomat for SDS.

Eg: Excel.

Q. How to DEVELOP a SDS database ?

sol- The steps to DEVELOP a SDS database are -

1. DEFINE, the OBJECTIVES of the system.
2. DEFINE, the COVERAGE of the system.
3. OBTAIN, the NT DATA of work Elements.
4. CLASSIFY, the WORK ELEMENTS.
5. ANALYZE, the data to determine ELEMENTAL NORMAL TIMES.
6. DEVELOP, the DATABASE to predict the NT values.
7. PREPARE DOCUMENTION.

⟹ WORK ELEMENT CLASSIFICATION in SDS -

- SDS is ORGANISED by Work Elements.
- Here the value of NORMAL TIME is a PREDICTED value rather than an OBSERVED value.
- The types of WE include :

① SETUP & PRODUCTION Elements

② CONSTANT & VARIABLE Elements.

③ WORKER - PACED & MACHINE Elements

④ REGULAR & IRREGULAR Elements.

⑤ INTERNAL & EXTERNAL Elements

↳ ① Setup & Production Elements -

- The CHANGOVER b/w Prodⁿ Runs is f/s "SETUP Time" or "CHANGEOVER TIME".
- Setup Time is a LOST PRODⁿ Time & it is the major disadvantage of Batch Production.
- Setup Elements = It is the activities associated to accomplish the changeover.
- Production Element - It is the activites associated with the processing of work units.
- Setup Element occurs Once per BATCH
- Prodⁿ Element occurs once per WORK UNIT.

∴ Total Batch Time (T_b) = $T_{su} + \Sigma T_c$

↳ (2) CONSTANT & VARIBLE Elements.

- CONSTANT Elements have the SAME TIME value in all Time Studies.

↳ Eg's of Constant Elements are -

(1) Replace Cutting Tool in Tool Post

(2) Dial Telephone number of customers

(3) Push 2 buttons to initate press stroke.

- VARIBLE Elements have the same BASIC MOTION Patterns & the same BASIC fns b/w jobs, but the NORMAL TIME varies because of differences in work variables.

- NORMAL TIME for Variable Element depend upon JOB characteristics.

↳ Eg's of Variable Element are -

(1) Keypunch Addres - Depends on -
(i) source of data
(ii) No. of lines
(iii) Total no. of Character.

(2) Fetch Kit of components for assembly -

(3) Load workpiece into Lathe & tighten Chuck.

(4) Turn workpiece in 1 pass.

- For Variable Elements. the JOB for the analyst is 2 Folded.

ie

(1) Identify the factor / features that govern the Normal Times

(2) Determine the exact nature of this relationship, so as to use it as a Predicting model for future performances.

↳(3) Operator Paced & Machine Controlled Elements -

- Operator Paced Element are MANUAL Elements involving the use of a TOOL
- It includes both SETUP & PROD^n elements.
- They can be CONSTANT or VARIBLE Elements.
- They includ RANDOM & UNAVOIDABLE variations.

- Machine Controlled Element - They are usually VARIBLE Elements & depend upon :

  (i) Operating Parameters

  (ii) Characteristics of Work Unit.

Eg: (1) Turn workpieces in One Pass.

↳ (4) REGULAR & IRREGULAR Elements :-

- Regular Element - • It occurs at LEAST once every work cycle.
  • It can be CONSTANT or VARIBLE PACED.
  • It can be operated Paced or Machine - Controlled Element.

Note - Regular Element = Prodⁿ Element

- Irregular Elements - • It occurs LESS FREQUENTLY than Once per work cycle.
  • It is PRORATED into a Regular Element.

↳ (5) External & Internal Elements -

- These types of Elements are largely observed for -
  - Regular Element
  - Production Elements
  - Machine Controlled Elements.

- External Elements - • These are Elements performed in Series ∈ Machine Time.

  • They are INCLUDED in Calc of Standards.

- Internal Elements - • These are Elements performed in Parallel ∈ the Machine Time.

⟹ ADVANTAGES of SDS -

(1) PRODUCTIVITY IN SETTING STANDARDS

(2) COST SAVINGS - for Large no. of Time Stds.

(3) SETTING STANDARDS BEFORE PRODUCTION

(4) AVOIDS PERFORMANCE RATING

(5) CONSISTENCY IN STANDARDS

⑥ INPUTS TO OTHER INFOMATION SYSTEMS -

- Time Values Can be also used for -
  - (i) Product Cost Estimation
  - (ii) Material Requirement Planning (MRP)
  - (iii) Enterprise Resource Planning (ERP)

⇒ Limitations (or) Disadvantages of SDS -

1. High Investment Cost.
2. Source of Data
3. Methods Description
4. Risk of Improper Application.

(II) Work Sampling -

Def - It is the statistical technique for determining the proportions of time spent by subject in various categories.

- For Work Sampling, usually -
  1. Large no. of observations are made.
  2. Obs. are made over extended periods
  3. For statistical Accuracy, the obs. are taken at RANDOM.

↳ Situations that warrant / demand the need of W.S (or) General characteristics of work situations where W.S is needed.

- (a) Sufficient time available for study
- (b) Multiple subjects
- (c) Long cycle Times
- (d) Non-Repeatative work cycles.

↳ Other names for Work Sampling -

Work Sampling can also be reffered to as -

(i) Activity Sampling

(ii) Occurance Sampling

(iii) RATIO DELAY STUDY

Note -

Work Sampling was Introduced by :

" L.H.C. Tippett " & LHC Tippet called W.S as " SNAP READING METHOD."

Then - R.L. Morrow changed the name to -

" RATIO DELAY Study "

Then - C.L. Brisley & H.L. Waddell coined the term " Work Sampling "

⟹ How to do Work Sampling -

↳ Applications of W.S (or) Parameters that specify the req. for W.S are -

1. Machine Utilization
2. Worker Utilization
3. Allowances for Time Standards
4. Average Unit Time
5. Time Standards.
   - ↳ Usually includes Indirect Labour & office work.
   - ↳ Should not be used for Incentive Pay Work due to Statistical Inaccuracy.

Problem 1: Study Eg 1:

Example 1 Work Sampling to Study Machine Utilization

A total of 500 observations were made at random times during a one-week period (a total of 40 hours) in the automatic lathe section. The subjects consisted of 10 identical machines. In each observation, the activity of the machine was identified and classified into one of three categories: (1) being set up, (2) running production, or (3) idle. The number of observations in each category is summarized in the table below. Estimate how many hours per week an average machine spent in each of the three categories.

Category	Number of Observations
(1) Being set up	75
(2) Running production	300
(3) Machine idle	125
	500

Solution: The proportion of observations spent in the setup is 75/500 = 0.15, the proportion spent in production is 300/500 = 0.60, and the proportion spent idle is 125/500 = 0.25. Based on a 40-hour week, these proportions convert into the hourly values per week for each machine given in the table below.

Category	Proportion of Observations		Hours per Category
(1) Being set up	0.15 × 40	=	6
(2) Running production	0.60 × 40	=	24
(3) Machine idle	0.25 × 40	=	10
	1.00 × 40	=	40

1. Make the Proportions.
2. Multiply the proportion to the Total no. of hrs.

⇒ Statistical Basis of Work Sampling -

- The statistical basis of Work Sampling is the BINOMIAL DISTRIBUTION.

where - p = TRUE proportion of Time spent engaged in a activity.

∴ for 'K' different activities / categories -

then - $$p_1 + p_2 + p_3 + \dots + p_k = 1 \text{ (Unity)}$$

then - μ = Mean

ie $$\mu = \text{mean} = n \cdot p$$

here n = Total no of observations

p = True proportion.

σ = Standard deviation.

ie $$\sigma = \sqrt{np(1-p)}$$

The parameters of μ & σ are CONVERTED BACK to Proportional Values as -

(i) $$\mu = \text{mean} \Rightarrow \quad p = \frac{\mu}{n} = \frac{pn}{n}$$

(ii) $$\sigma_p = \frac{\sqrt{np(1-p)}}{n} = \sqrt{\frac{p(1-p)}{n}}$$

where σ_p = Std. Dev of Prop: 'p'

Then -

$\hat{p}$ = Proportion of Total no. of Observations

Here $\hat{p}$ must be of LOW VARIANCE & ABSENT of BIAS.

↳ What is Biased Estimate ?

Def: Biased Estimate is the one that tends to differ CONSISTENTLY from the True Value

This might happen due to -

(i) Estimating Method maybe FLAWED.

(ii) Variables estimated are influenced due to OBSERVING it.

Thus the BIAS for $\hat{p}$ is reduced by =>

"RANDOMIZING the OBSERVATION times."

↳ what is LOW VARIENCE ? -

Def - It means the variable of interest is measured multiple times & all of the measured values will be close together.

Variance = $\sigma_{\hat{p}}^2$

→ (1) Confidence Interval in W.S.-

Note - We expect $\hat{p} \simeq p$

Here -

$$P_r\left(\hat{p} - Z_{\alpha/2} \cdot \hat{\sigma}_p < p < \hat{p} + Z_{\alpha/2}\, \hat{\sigma}_p\right) = 1-\alpha$$

"Probability of"

where $\hat{\sigma}_p$ = Proportion of Standard Deviation.

ie The above eq^n expresses the "CONFIDENCE INTERVAL" as - the value of p lying b/w the intervals - $(\hat{p} - Z_{\alpha/2} \cdot \hat{\sigma}_p)$ & $(\hat{p} + Z_{\alpha/2} \cdot \hat{\sigma}_p)$ & at a "CONFIDENCE LEVEL" of "$(1-\alpha)$".

↳ Typical Values of CONFIDENCE LEVELS to remember -

(1) $Z_{\alpha/2} = 1.65$ for 90% confidence level.

(2) $Z_{\alpha/2} = 1.96$ for 95% confidence level.

Problem 2 - Eg 2 :

Example 2 Confidence Interval in Work Sampling

Determine the 95% confidence interval for the proportion of time spent setting up the machines, category (1), in Example 1.

Solution: From the solution of Example 1, we know that the proportion of time spent setting up machines is $\hat{p} = 0.15$. Computing the standard deviation using equation (4), given that there are a total of 500 observations ($n = 500$),

$\hat{\sigma}_p = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ ← $\hat{\sigma}_p = \sqrt{\frac{0.15(1 - 0.15)}{500}} = 0.01597$

$q = 1 - p$

For the 95% confidence level, the corresponding $z_{\alpha/2} = 1.96$; thus

$$\hat{p} - z_{\alpha/2}\hat{\sigma}_p = 0.15 - 1.96(0.01597) = 0.15 - 0.01313 = 0.1187$$

and

$$\hat{p} + z_{\alpha/2}\hat{\sigma}_p = 015 + 1.96(0.01597) = 0.15 + 0.013 = 0.1813$$

Accordingly, we can state that at the 95% confidence level, the true value of the setup time proportion lies between 0.1187 and 0.1813. → Required Interval.

→ (2) No. of Observation Required in W.S -

Here Statistical Errors can be REDUCED by INCREASING the no. of OBS.

In order to decide the no. of OBS to be made -

(1) Define the confidence Level (i.e (1 - α))

∴ Consider " c " as " INTERVAL PARAMETER"

Here 'c' is also a PROPORTION like 'p'

∴ The confidence level is taken as " p ± c "

where -

$$c = Z_{\alpha/2} \cdot \hat{\sigma}_{\hat{p}}$$

WKT-

$$\hat{\sigma}_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\therefore \quad \hat{\sigma}_{\hat{p}}^2 = \frac{\hat{p}(1-\hat{p})}{n}$$

$$\therefore \quad n = \frac{\hat{p}(1-\hat{p})}{\hat{\sigma}_{\hat{p}}^2} = \frac{(Z_{\alpha/2})^2 \hat{p}(1-\hat{p})}{c^2}$$

where n = Total no. of Observations

Problem 3 - Eg 3 :

Example 3 Determining the Number of Observations Required

In the context of our previous examples, determine how many observations will be required to estimate the proportion of time used to set up the 10 machines in the automatic lathe section. The confidence interval must be within ± 0.03 of the true proportion, which the foreman initially estimates to be $\hat{p} = 0.20$. A 95% confidence level will be used.

$\hat{p} \pm c$

Solution: At the 95% confidence level, $z_{\alpha/2} = 1.96$. The confidence interval is 0.20 ± 0.03, so the half-width of the interval c is 0.03. From Equation (8),

$$\hat{\sigma}_{\hat{p}} = 0.03/1.96 = 0.0153 \qquad \therefore \hat{\sigma}_{\hat{p}} = \frac{c}{Z_{\alpha/2}}$$

Using this value in equation (9), we obtain the required number of observations:

$n = \frac{\hat{p}(1-\hat{p})}{\hat{\sigma}_{\hat{p}}^2}$ ↑

$$n = \frac{0.20(1-0.20)}{(0.0153)^2} = 683.5, \text{ rounded up to 684 observations.} \quad \blacksquare$$

Equations (8) and (9) can be combined to compute n in one step rather than the two steps used in Example 3:

Comb. of 2 formula

$$n = \frac{(z_{\alpha/2})^2 \hat{p}(1-\hat{p})}{c^2} \qquad (10)$$

→ ③ Determining Average Task Time & Time Standards –

- Time Standards set by W.S are not accurate

ⓐ Average Task Time :-

Def :- It is the amount of time required to complete an average work unit. & Σ no consideration of the Worker's performance.

ie $$T_{ci} = \frac{p_i (TT)}{q_i}$$

where T_{ci} = Average Cycle Time for task i

ie T_{ci} = Average Task Time.

TT = Total Time of W.S (Hrs / min)

q_i = Total quantity of units produced

p_i = Proportion of Observations associated with the work category.

Example 4 Determining Average Task Times in Work Sampling

Suppose in Example 1 that a total of 1572 work units were completed by the 10 machines and that a total of 23 setups were accomplished during the 5-day period. Determine (a) the average task time per work unit during production and (b) the average setup time.

Sol:- Here Q_i = 1572 units

Here Total Time for W.S = No. of M/c × No. of Hours

$TT = 10 \times 5 \times 8$

$TT = 400$ hrs

(a) WKT- $T_{ci} = \frac{\hat{p}_i (TT)}{Q_i}$

Here 1572 units are produced under the category "Production"

WKT - $\hat{p}_i = 0.60$

$\therefore T_{ci} = \frac{0.60\,(400)}{1572} = 9.16$ min

(b) No. of setups - Q_{su} = 23. & $\hat{p} = 0.15$

$\therefore T_{su} = \frac{0.15(400)}{23} = 156.52$ min/setup

(b) For Time Standards -

- W.S is most suited for -
  1. INDIRECT LABOUR ACTIVITY
  2. CLERICAL OFFICE WORK
- Here the variability may be caused due to -
  - (a) Irregular Elements
  - (b) Adjustments / Testings
  - (c) Mental Analysis.
- How to set time Standards ?

1. Analyst must record the "Rate of Performance" during each observation & also identify the Category of Activity.
2. For Non-Work activities the Performance Rating is 0.
3. Then NORMAL TIME is determined as -

$$T_{n_i} = \frac{P_i \, (TT) \, (\overline{PR_i})}{Q_i}$$

where T_{n_i} = NT of a Category 'i'

$\overline{PR_i}$ = Avg. Performance Rating (decimal fraction)

④ Then the Standard Time is estimated by including certain allowances -

ie $T_{std_i} = T_{n_i} (1 + Apfd)$

→ General Steps for conducting a Work Sampling Study -

① Define the OBJECTIVES of the Study.

② Define the SUBJECTS to be studied

③ Define the OUTPUT MEASURE.

④ Define the ACTIVITY CATEGORIES.

⑤ DESIGN the Study.

⑥ IDENTIFY the Observer / Analyst.

⑦ ANNOUNCE the study

⑧ Make the OBSERVATIONS.

⑨ After completion, STUDY, ANALYSE & PRESENT the results.

⟹ APPLICATION Issues in Work Sampling -

- W.S study is CUSTOM-DESIGNED.

But the IMPLEMENTATION issues are -

1. DEFINING the ACTIVITY CATEGORY
2. WORK SAMPLING FORMS
3. SCHEDULING the OBSERVATION TIMES.

(1): DEFINING the ACTIVITY CATEGORY -

- Activity Categories are the states of the subject that the OBSERVER must IDENTIFY & CLASSIFY during his Observations.
- Initially the no. of Categories must be less (about 10).
- Fewer no. of Categories $\Rightarrow \hat{p}_K \uparrow$. & this reduces the variance & improves the confidence levels.

presentative of actual studies.

TABLE 2 Possible Activity Categories for a Work Sampling Study on Machinists in a Small Lot Job Shop

Category	Description
1. Plan	Worker is studying engineering drawing in preparation for new job.
2. Setup	Worker is setting up machine tool for new job.
3. Run production	Worker is running machine tool in production of parts.
4. Cleanup	Worker is cleaning up machine tool and immediate work area.
5. Wait	Worker is waiting for materials, tools, instructions from foreman, etc.
6. Personal	Worker is taking personal time, rest break, or coffee break, conversing with coworker.
7. Away	Worker is not at machine, not in sight.

TABLE 3 Possible Activity Categories for a Work Sampling Study on Office Workers

Category	Description
1. Keypunching at computer terminal	Each desk is equipped with a personal computer. Worker is involved with the operation of a computer.
2. Writing at desk	Worker is filling out paper forms, or engaged in similar paperwork.
3. Filing	Worker is at central file area, performing either storage or retrieval tasks.
4. Telephone	Worker is using telephone.
5. Walking	Worker is walking within office area.
6. Conversing	Worker is conversing with coworker, supervisor, or customer in office.
7. Personal	Worker is taking personal time, rest break, coffee break, etc.
8. Away	Worker is not at desk, not in office.

(2): WORK SAMPLING FORMS -

- The forms used in the study for the observations must be designed specifically for that study.
- This is due to the reason that -

Act Categories, Subjects, Total no. Obs. time periods vary b/w Studies.

Q. what are the Objectives in the Design of W.S forms ?

Sol- Objectives are -

① Make the form EASY & CONVIENT to use.

② Taylor made to making SNAP READINGS & the facilate the QUICK recording of data.

③ : SCHEDULING OBSERVATION TIMES :-

- For a W.S study, the no. of obs. is LARGE.
- It can further be Classified as -

ⓐ PREPARING a SCHEDULE of RANDOMIZIED OBSERVATION Times :-

- Observation times must be RANDOM so as to improve STASTICAL accuracy.
- An alternative for Scheduling Observation times is to use the Principle of "Sampling Stratification."

Def of Sampling Stratification :

"It is a technique in which the total no. of observations is ÷ into a specified No. of time periods (ie days, wks etc). Such that there are EQUAL no. of samples taken in each period.

- Sampling Stratification is known to REDUCE the Variance in the Estimated proportion.
- Stratification is more convenient for observers to use.
- Stratification adds an Element of STRUCTURE to the Study.

↳ Advantages of Sampling Stratification:

1. Reduces Variance
2. Easy for Analyst to use
3. Adds an Element of STRUCTURE.

↳ What is the actual procedure of Randomization?

The following sources can be used to obtain randomized times :-

(a) Handbooks / Textbooks

(b) PSEUDO-RANDOM number generators on Spreadsheet softwares.

(c) PSEUDO-RANDOM number generators on Engineering Calculators

(d) Last few Pages of a Telephone Directory.

- Whatever be the source, the nos obtained as finally converted to clock time in the period of interest.

(b) PERSONAL DIGITAL ASSISTANTS & RANDOM TIME BEEPERS.

Note - An advantage of W.S study is the analyst could be engaged in other duties during the W.S study duration.

- If the analyst always looks at the clock so as not to miss the Scheduled Sampling Times, then the analyst cannot devote the attention to other tasks.

- Hence few devices like -

  (i) Personal Digital Assistant (PDA's)

  (ii) JD-7 Random Reminder

↳ Application of PDA & JD-7 Random Reminder ⇒.

- PDA : are SMALL PERSONAL COMPUTERS
- JD-7 Random Reminder : It is sold by "Divilbiss Electronics." It is a LIGHT WEIGHT box that can be clipped to one's BELT.
- These type of devices can be used to serve 2 FUNCTIONS in a WORK SAMPLING study:-

(a) Random Timer : It helps to schedule random times for the Obs. runs.

- The analyst sets a "SAMPLING RATE PARAMETER"
- Sampling Time Intervals can range from several minutes to several Hours.

(b) Beeper : • With a set Sampling Rate, the device provides either a hearable beep for 1 second (or) Vibrating Signal lasting for several seconds.

↳ Advantages of Work Sampling -

(a) Multiple Subjects can be included.

(b) It costs LESSER than direct observation techniques

(c) Reduces SHORT TIME observations in the work routine. (techniques such as CONTROL CHARTS can be used to identify Abberations).

(d) Less Training Requirement

(e) It is Less TIRESOME & TEDIOUS.

(f) Operations which are too costly to observe Directly, can be OBSERVED.

(g) It is less DEMANDING as a subject in the study.

↳ LIMITATIONS or Disadvantages of Work Sampling -

(a) Due to lack of Accuracy, it must not be used to set Time Standards.

(b) It must not be used for Incentive Payment Plans.

(c) It is not ECONOMICAL to study a single subject.

(d) It is not SUITED for subjects placed at a large DISTANCE.

(e) It provides LESS INFO. on the work Elements.

(f) Since it is performed on many subjects. it tends to AVERAGE their activities.

(g) Since it is based on STATISTICAL THEORY, it is NOT easily understood by the workers.

(h) If the method of the worker changes then due to INSUFFICIENT DOCUMENTATION, the result may become irrelevant.

(i) The observation of the subject may influence their behaviour & thus provide incorrect conclusions.

III LEARNING CURVES -

Def: The Learning Curve Phenomenon refers to the reduction in cycle Time that occurs in a repetative work activity as the no. of cycles increases.

ie "Pratice makes Perfect."

- The Learning Effect occurs @ diminishing rates as the no. of Repeations Increase.
- LEARNING Curves are also -

  "MANUFACTURING PROCESS FUNCTION" &

  "EXPERIENCE CURVE."

- Learning Curve effects CYCLE TIME & the Reduction in PRODUCT COST.

→ Learning Curve Theory :-

- It says "There is a CONSTANT LEARNING RATE that Applies to a given Repetative Task".

Q. What is Learning Rate? (aka Doubling Effect)

Sol: It is defined as the Proportion by which the dependent variable (ie Task Time) is multiplied

every time the no. of work cycles DOUBLES.

Q. What is the RATE of IMPROVEMENT?

Sol- It is the proportion by which the dependent variable is reduced every time the no. of units DOUBLES.

Note - IR = Improvement Rate

LR = Learning Rate

Here IR & LR are complemented to each other as-

$$IR + LR = 1$$

→ History of Learning Theory -

- It was introduced in the seminal paper titled- "Factors Affecting the Cost of Airplanes" which was given by T. P. Wright.
- The other important Log-Linear model was given by J. R. Crawford.

→ LEARNING:

- Learning is NOT the same for everyone.
- Physical, Psychological, Environmental, & many other factors affect the learning.

Q. What are the roles of an IE as a Learner?

Ans: Roles include -

① Is to Learn the Process

② To ensure proper COMPATIBILITY with the work Environment & ensure there is NO BIAS.

③ To observe & make justified observation to retain graphical accuracy.

⇒ Why does the Learning Curve Occur? -

- Here the Worker's contribution is Limited.

Q. How can worker Reduce the Cycle Time by ↑ the no. of Units produced?

Ans: Some factors are:

① Worker is FAMILIAR ε the Task.

② Worker makes FEWER MISTAKES.

(3) Worker's HAND & BODY movements become more EFFCIENT. & a RHYTHM is developed etc.

Q. What is CONTINOUS IMPROVEMENT?

Ans: It is sometimes used to refer to "the constant search for ways to reduce the cycle time & costs. & to improve the quality.

Q. What are the contributions to Learning phenomenon by the ORGANISATION?

Ans: The other factors include:

(1) Method's improvement is done by an IEM department.

(2) Fine - Turning of Machinary

(3) Developement of Special Tools

(4) Technological improvements in Processes. Tooling & Mechinary.

(5) Product Design improvement

(6) Improved Quality of Incoming material.

(7) Better planing & scheduling

(8) Imp Logistical Support.

⟹ Determining the Learning Rate -

- For the LOG-LINEAR learning curve, the most important information is the Learning Rate.
- Here the LR is more of a GENERIC PARAMETER

Q. What are the method that are used to determine the learning rate?

Ans: There are 2 methods, which are -

ⓐ USE INDUSTRY AVERAGES

ⓑ ANALYZE DATA from the ACTUAL APPLICATION.

↳ ① Use Industry Averages:

- The following tables has few commonly listed values of the LR.

TABLE 2 Typical Learning Rates (Crawford Model) for Various Types of Work and Industry Categories

Type of Work	Learning Rate (%)	Industry Category	Learning Rate (%)
Assembly, electrical harness	85	Aerospace	85
Assembly, electronics	85	Complex machine tools	75 to 85
Assembly, mechanical	84	Construction	70 to 90
Assembly of prototypes	65	Electronics manufacturing, repetitive	90 to 95
Clerical operations, repetitive	75–85	Machine shop, repetitive	90 to 95
Inspection	86	Punch press operations, repetitive	90 to 95
Machining, repetitive	90	Purchased parts	85 to 88
Sheet metal work	90	Raw materials	93 to 96
Welding	85–90	Shipbuilding	80 to 85

- Other factors also effect these rates, such as influences by:
  - (a) Management
  - (b) Labour Unions
  - (c) Cultures
  - (d) Customs etc.

↳ (2) Using Data from the Application:

- By this method the LR can be determined by the following techniques.
  - (a) OBSERVING the DOUBLING EFFECT
  - (b) FINDING the SLOPE from any 2 Obs.
  - (c) REGRESSION ANALYSIS using MULTIPLE OBSERVATIONS

↳ Regression Analysis - (also Least Squares curve-fitting Procedure)

- The other 2 Techniques only deal ō 2 Observations.

⟹ FACTORS affecting the LEARNING CURVE -

Q. What are the factors affecting / influencing the learning curve?

PTO.

Sol- "Type of Work" & "Industry Category" are NOT the only factors.

The other factor influencing the LR are:-

1. Learning in Worker - Machine Systems
2. Composite Learning Curves
3. Interruptions in the Learning curves
4. Product Complexity
5. Preproduction Planning
6. Labour Turnover
7. The Plateau Model.

⇒ ① Learning in Worker - Machine System -

- Here the Worker's Manual Portion is affected by the Learning curve phenomenon

⇒ ② Composite Learning curves -

Def: It refers to the Situation in which a learning rate is to be determined for a large entity of work.

- This procedure was developed by Mc Campbell & Mc Queen.

⇒ (3) Interruptions in the Learning Curve -

- Here the reasons of the interruptions may be -

(i) Batch Production :
- Here a given part is manufactured in a specified order quantity & there is a Time delay until the next order of the same part.

(ii) Labour Strikes

(iii) Vacations

(iv) Raw Material Shortage.

Q. What is REMISSION ?

Ans: It is the LEARNING LOSS, that shows up as an UPWARD STEP in the cycle time.

REMISSION : It the learning loss, caused to INTERRUPTIONS in a CONTINOUS PRODUCTION.

(Or)

REMISSION: It can be defined as the INCREASE in Cycle Time b/w N & N+1 units

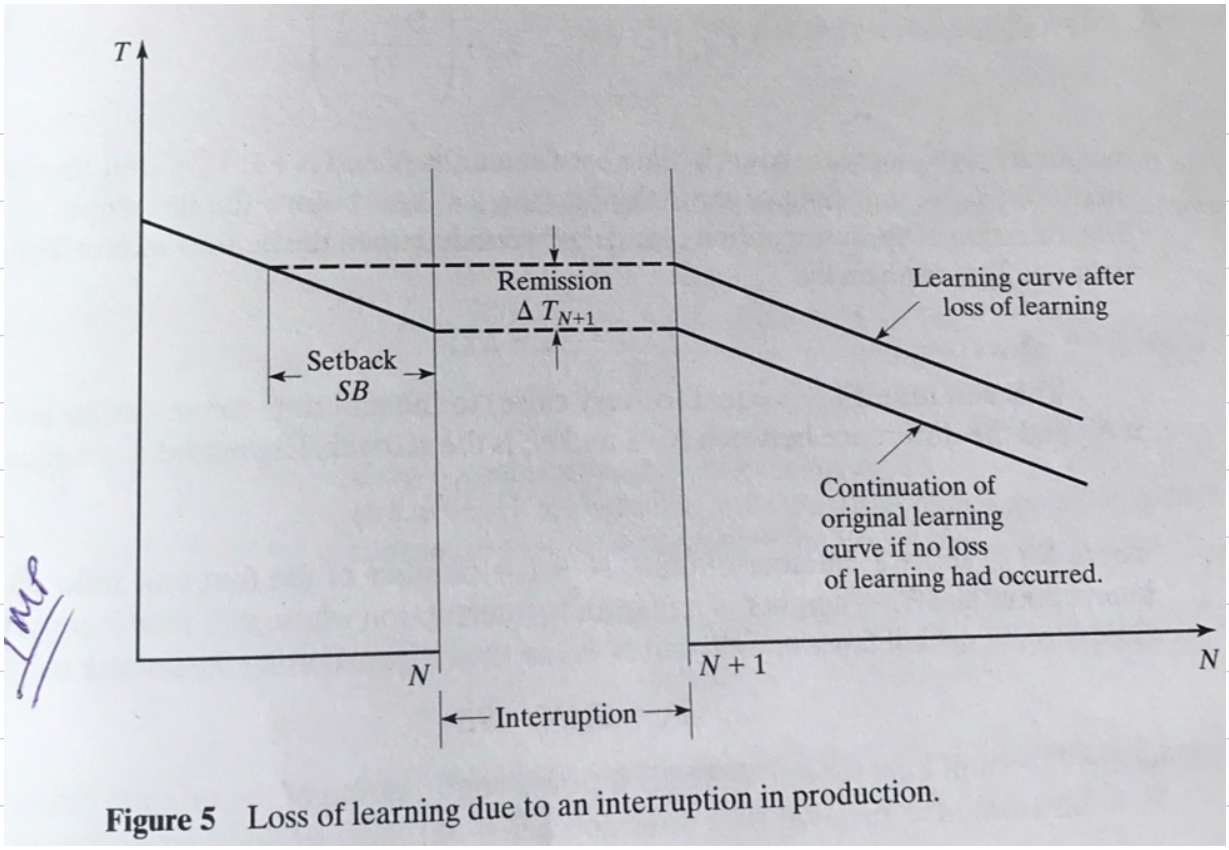


Figure 5 Loss of learning due to an interruption in production.

Q. What are the Reasons for LOSS OF LEARNING during INTERRUPTIONS ?

Sol- The reasons include :

1. The worker might FORGET the previously learned information.
2. NEW WORKERS are SUBSTITUTED for old ones.
3. The Standard Description for the method is LOST.
4. The supervision FORGETS the SUBTLE TRICKS which may not be mentioned in the method's description.
5. Special Tooling is LOST or forgotten.
6. Workplace is not Efficiently Laid out.

Q. What is meant by SETBACK?

Sol- SETBACK is defined as: The no. of units lost along the N-Scale.

⇒ (4) Product Complexity :-

- It is measured by factors such as the no. of components in a final assembly or the direct labour time content in fabricating the product.
- Higher the Complexity ⇒ More is the learning.
- If the time to complete the 1st unit is high, then there is a great chance to reduce the proportion of time in subsequent units.

⇒ (5) Preproduction Planning:

- If the amount of "PREPRODUCTION PLANNING" is HIGH, then the TIME to Complet the 1st unit, will be depressed. & the slope 'm' will be flatter.

Q. What does Preproduction Planning include?

Ans: (1) Tooling (2) Machine Selection DFM DFA

(3) Methods Analysis (4) Product Design

(5) Worker Traing (6) Scheduling & Logistics support.

⇒ ⑥ Labour Turnover:

- Bringing in Replacements of Older Workings disrupts LEARNING.
- It is seen as ⇒ ↓ IR & ↑ LR.

⇒ ⑦ The Plateau Model -

- It refers to the belief that "LEARNING EVENTUALLY CEASES".
- Here there are 2 phases:

  ⓐ Start-up phase: Here the learning progresses according to a log-linear fn.

  ⓑ Steady-state phase: Here no further improvement in the dependent variables occurs.

Q. What are the reason that the Steady-State phase is seen?

Ans: It includes mainly 4 reasons which are:

① Worker is limited due to the SPEED of the Equipment.

② Workers are working @ a higher rate than Standard performance, but are NOT PAID for the same.

③ Management is unwilling to INVEST in newer / more productive Technologies.

④ Management believes that the Learning Curve CANNOT CONTINUE INDEFINITELY.

⇒ Applications of the Learning Curves -

① Time Standards vs Learning Curve.

Q. What are the conditions that ensure Time Standards are of constant values?

Ans: The following conditions are:

① Method used is Standardized

② Work units are Standardized.

③ It is performed by an Average Worker with the Standard Performance Pace.

Q. How is it that the Standard time remains constant with doubling of work units & a decrease in LR in every subsequent unit?

Ans :- Here the Standard Time = Constant.

- But Actual Cycle Time = Variable.
- Variation can occur due to the worker's effort in learning & becoming proficient in their jobs & thus reducing the cycle time.
- If the worker has improved their skill & job, then rewards such as "INCENTIVE PAY" must be given.
- However if the reduction/variation occurs due to "changes in the Method / work unit".
- Then these changes are to be incorporated by "Changing the defenation of the Standard Method".
- Overall, the METHOD & WORK UNITS are responsible for Time Standards.

Q. Can the worker "Beat the System" & how?

Ans:- If the worker has found an improved way to work, then by not notifying the higher authorities, the worker has "Beat the System".

Q. What is Standard Reference Quantity (SRQ)?

Ans: It is the Discrete Quantity for Production for which the Time Standard for a task is Applicable.

(Or)

Ans: It the production quantity for which the standard is still valid.

UNIT-5

(I) Computerized Work Measurement & Standards Maintainence

(II) LEAN PRODUCTION.

(I) Computerized Work Measurements & Standards Maintenance.

- Work Measurement is a COSTLY overhead function for an Organization.
- The Computerized System can be applied to -
  - (a) DTS
  - (b) PMTS
  - (c) SDS
  - (d) WS.
- In DTS & WS they REQUIRE the Analyst to be present during the study.
- Note - A common limitation of DTS & WS is: "Time standards cannot be determined before the job is running!"
- Common trends nowadays focuses on PMTS & SDS & its combination for computerized systems.

OBJECTIVE of computerized systems is:
"To reduce the manual & clerical content of the problem solving techniques by performing tedious & time consuming computational chores.

⇒① Computerized Tools for DTS -

Q. What are the CONVENTIONAL PROCEDURE to find Time Standards by DTS?

Sol: The conventional procedure includes -

① DEFINE & DOCUMENT the method.

② DIVIDE the Task into Work Elements.

③ TIME the worker element & obtain the observed time.

④ RATE the worker's performance

⑤ APPLY Allowance & Compute the Standard Time.

- When the ABOVE procedure is implement ed out a computer, then several hours is req to conduct the study.

Note - Computer's assistance can be opted in steps ③, ④, ⑤:

- Steps ① & ② fundamentally REQUIRE the ANALYST presence.

∴ The mental effort of the ANALYST is Essential in methods defn. & work Element Identification.

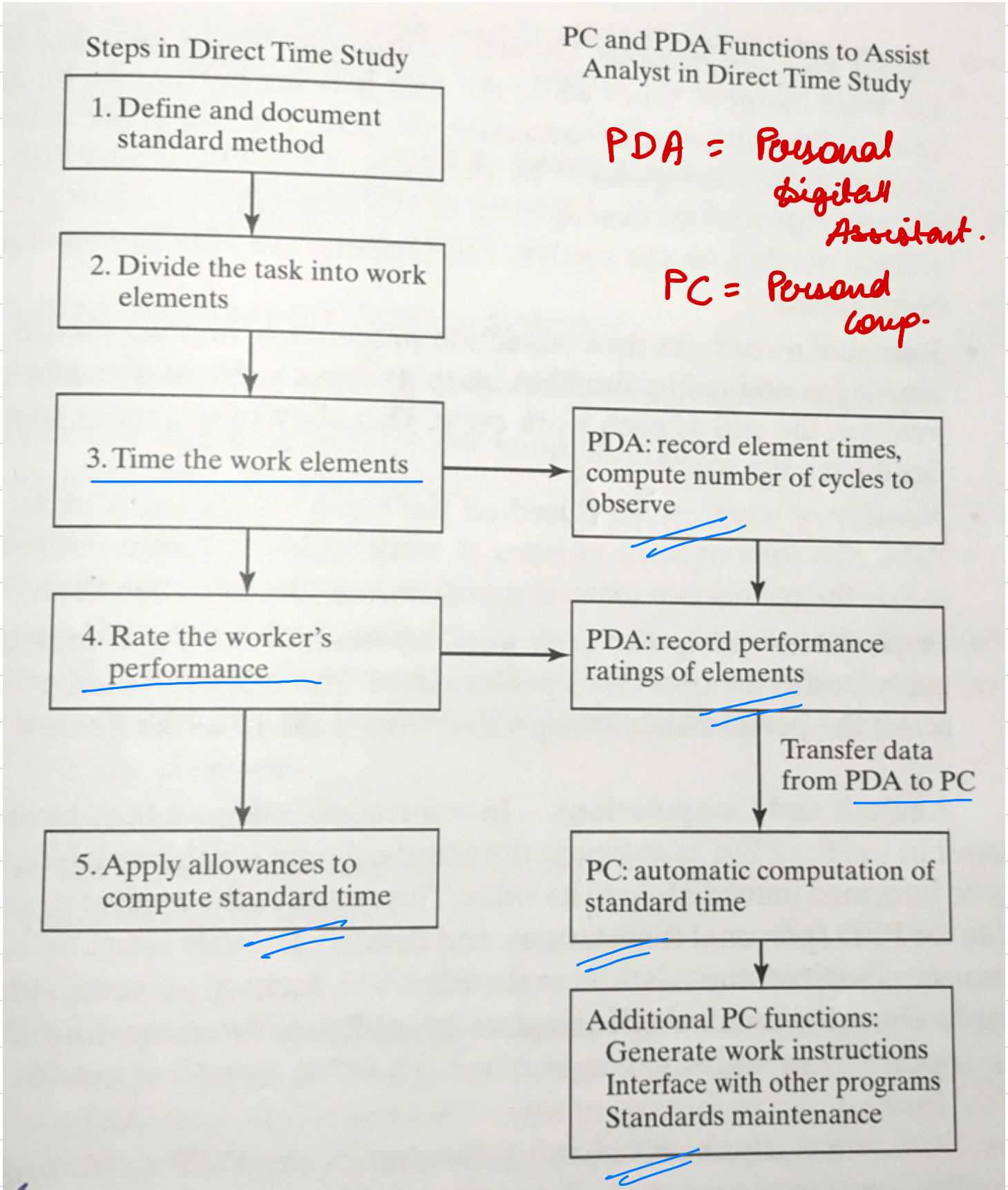


Figure 1 Steps in direct time study and how personal computers and personal digital assistants can be applied.

PDA = Personal Digital Assistant.

PC = Personal Comp.

VVV IMP Must Draw. Must Learn

⇒ Computer Assistance in steps ③, ④, ⑤ for DTS-

(a) COLECTING & RECORDING the Data -

- Here steps ③ & ④ are performed SIMULTANOUSLY.
- The PERSONAL DIGITAL ASSISTANT (PDA) Satisfies the requirement of a PORTABLE device.

Q. What are the ways in which Analyst use the PDA?

Ans: The following ways are:

(1) WORK ELEMENT DESCRIPTIONS: These include abbreviations of the elements defined in the standard method.

(2) TIMING the WORK ELEMENT: The PDA displays the revolving sequence of work elements & by tapping the screen the analyst is able to record & observe the time & work element occp.

(3) STASTITICAL RECORD: The PDA maintains a running record of the data & also computes the MEAN & STANDARD DEVIATION at the end of each cycle.

④ NUMBER OF CYCLES : The PDA also computes the req. no. of cycles to be observed for a given confidence interval.

⑤ PERFORMANCE RATING : The PDA allows the analyst to select the pace from a menu to the nearest 5%.

⑥ Analysis & computation :

- Here the Analyst transfers the data from PDA → PC & perform Analysis & hence compute the Standard Time.

Q. What are the OPPURTUNITIES & ADVANTAGES of using a PC for analysis & computation ?

Ans: ① STASTITICAL ANALYSIS & CHARTING - It involves feeding the data to spreadsheet softwares & to chart the data given.

② STANDARD TIME COMPUTATION - Here the Standard Time can be calculated in various units of Time.

③ GENERATION OF WORK INSTRUCTIONS :

④ INTERFACE TO OTHER PROGRAMS : The information can be shared to MRP, MRP II, CAPP, loading & ion scheduling softwares.

Here MRP = Material Requirements Planning

MRPII = Manufacturing Resource Planning.

CAPP = Computer Aided Process Planing.

(5) Standard Maintainence -

⇒ (2) Computerized Tools for Work Sampling -

Q. What is Procedure for work sampling using Computerized Tools?

Sol- The steps include :

(1) DEFINE the OBJECTIVE's of the W.S Study.

(2) DEFINE the SUBJECT to be Studied.

(3) DEFINE the OUTPUT MEASURE

(4) DEFINE the ACTIVITY CATEGORIES

(5) DESIGN the STUDY.

(6) IDENTIFY the OBSERVERS

(7) ANNOUNCE the Study.

(8) MAKE the Observation

(9) ANALYSE & REPORT RESULTS

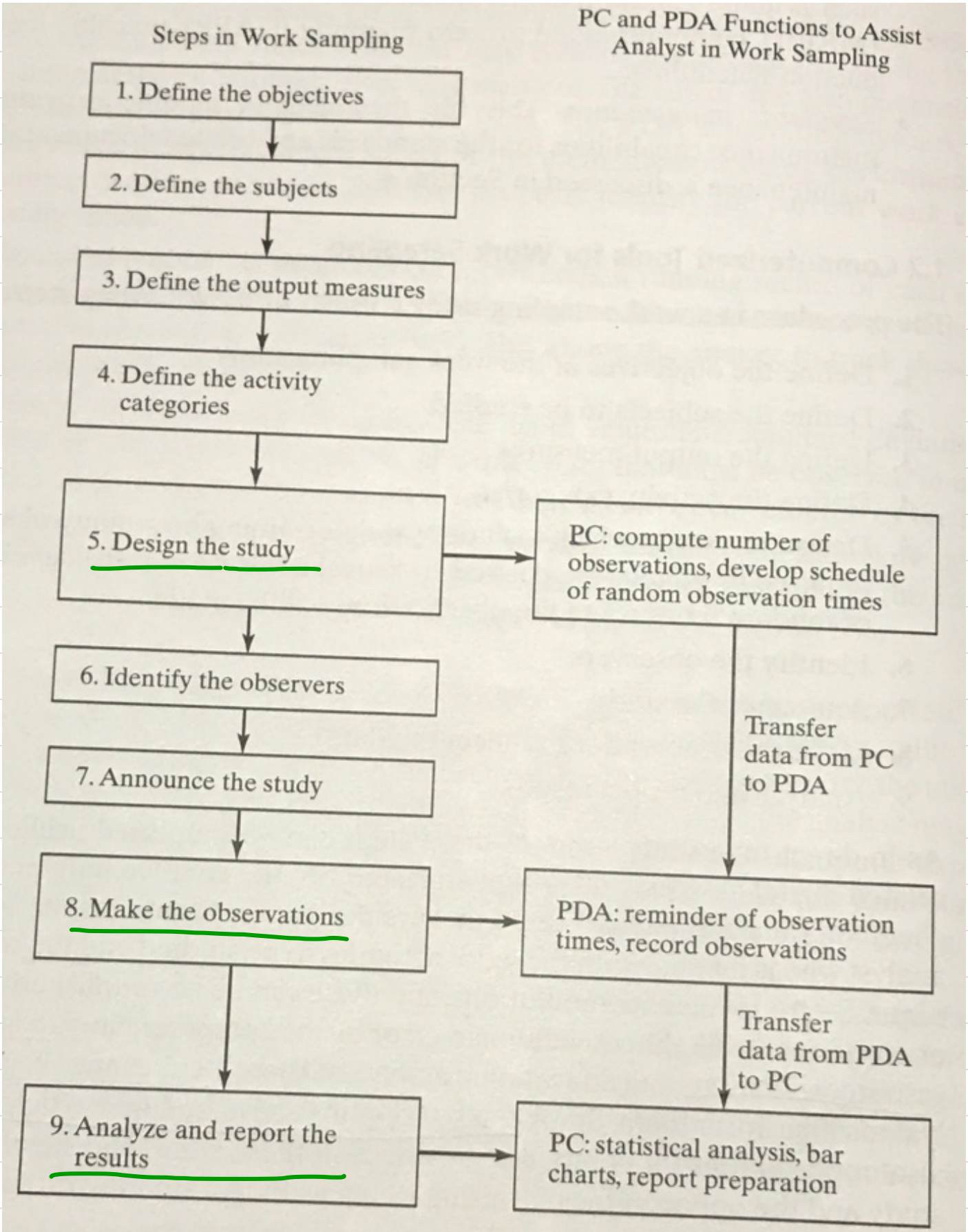


Figure 2 The steps in a work sampling study and the opportunities for using PCs and PDAs.

VVVVIMP. Must Draw. Must Learn

- Here Steps ①, ②, ③, ④ : Require the CREATIVE JUDGEMENT of the Analyst.
- Step ⑤ : Involves COMPUTATION & hence can be performed on a Computer.
- Step ⑥, ⑦ : It requires HUMAN RESOURCES DECISIONS.
- Step ⑧, ⑨ : Requires ACTIVE HUMAN participation but can be AUTOMATED.

∴ Steps ⑤, ⑧, ⑨ are computerized.

⟹ Computerized Assistance in Steps ⑤, ⑧, ⑨ -

ⓐ COMPUTATIONS DURING WORK SAMPLING DESIGNS -

- Work Sampling softwares are able to perform all the req. preliminary calculations during the designing of a work Sampling study.

Q. What are the capabilities of this software?

Ans: There are 2 capabilities :-

ⓐ Number of Observations → con. interval value?
→ con. level value?

ⓑ Schedule of Randomized Times.

Q. What are the specifications to be mentioned by the Analyst to create a Schedule of Randomized Times?

Ans: The specifications are:

1. NO. of OBSERVATIONS
2. No. of SUBJECTS
3. Period of STUDY
4. The SHIFT Schedule.

(b) Making the Observations:

- Traditionally it is a Paper & Pencil Activity

Q. What are the features of using a PDA for a Work Sampling method?

Ans: The features include:-

1. USING a RANDOM BEEPER - Alternatively the "JD-7 Random Minder" can also be used.
2. RECORDING the OBSERVATION DATA -
3. UPLOADING to a DESKTOP or PC.

(c) Analysis & Reporting of Results -

- The activity categories are often presented in a table that lists them in a DECREASING ORDER of their RELATIVE PROPORTIONS.

Q. How are the significance of the proportions emphasised ?

Ans: It can be emphasized by the following techniques-

(a) Use BAR CHARTS ⇒ Relative Percentages

(b) COMBINE activity categories into BROADER classes.

(c) Express the proportions in units of time / money & hence highlighting its importance.

(d) Chart the DAY-BY-DAY observation data on a Time Chart.

PTO.

⇒ (3) Computerized Tools for PMTS & SDS-

- Since PMTS & SDS are techniques of work measurement where the standards can be set while still in production. It is Ideal to use Computerized Tools to set these standards.

Q. What are the 2 principal functions of a computerized work measurement software?

Ans: The 2 principles fns are -

(a) Work Measurement fn -

(b) Standards Database fn. -

⇒ (a) WORK MEASUREMENT FUNCTIONS-

- The W.M fn requires the KNOWLEDGE & expertise of the Analyst.

" The Relation b/w Analyst & Computer is Always SYNERGISTIC "

PTO.

Q. What are the skills & Responsibilities undertaken by the Analyst?

Ans: (1) SKILLS :

- The analyst brings skills such as -

(a) ANALYTICAL SKILLS

(b) EXPERIENCE

(c) GOOD JUDGEMENT.

(2) RESPONSIBILITIES :

(i) DEFINE the STANDARD METHOD.

(ii) DECIDE as to what Tools & Equipment is required

(iii) DEFINE & EVALUATE the Work VARIABLES.

(iv) RETRIEVE data from Database.

(v) INTERACT ⊂ the SYSTEM.

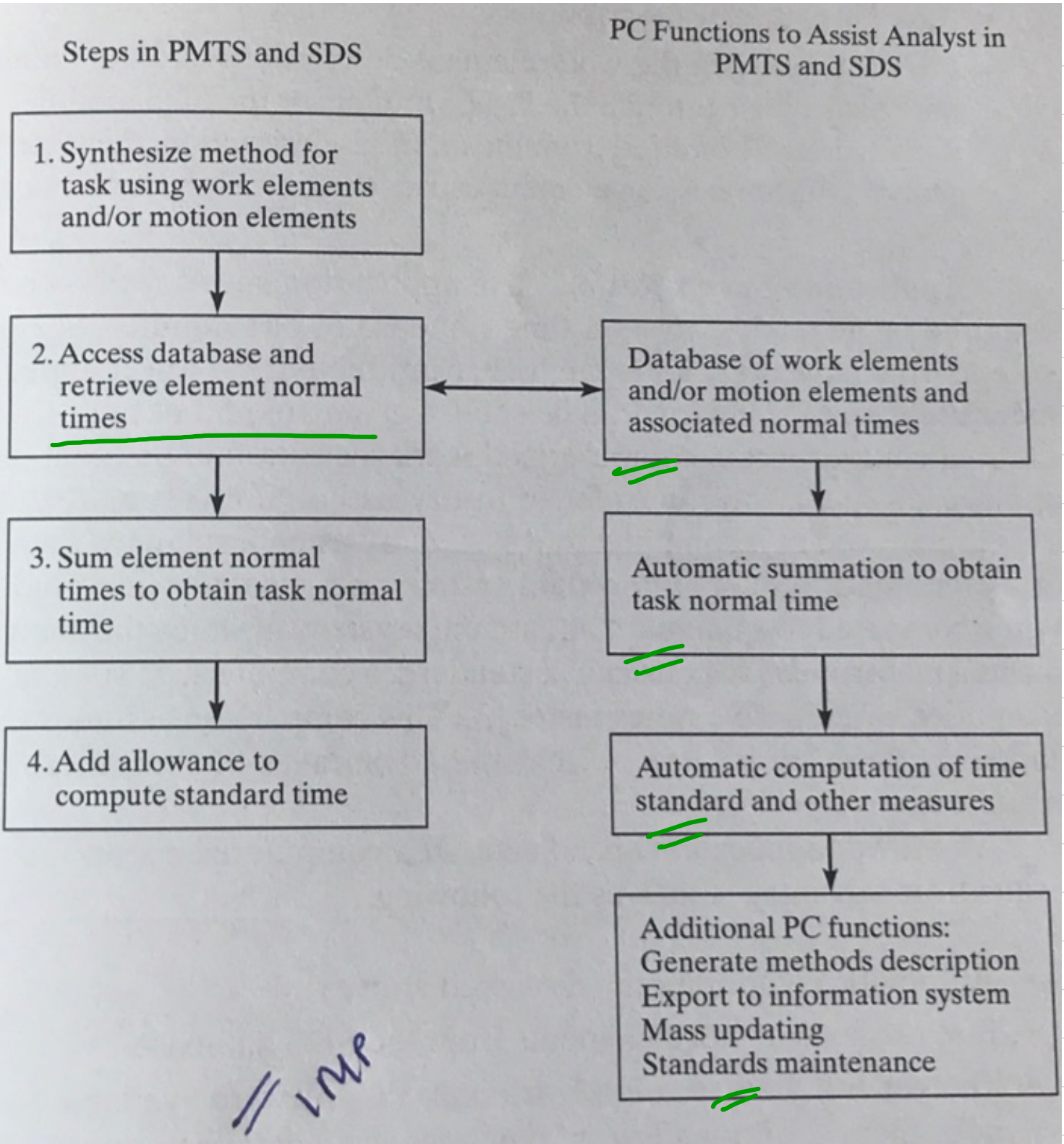


Figure 3 The steps in a PMTS and SDS work measurement system and the role played by a PC in the system.

MUST STUDY, MUST WRITE.

Q. Mention & discuss the FEATURES & ISSUE that are Associated with the WORK MEASUREMENT functions?

Ans: (i) AUTOMATIC COMPUTATIONS -

- It must perform all the necessary computations
- It can be used for a PMTS, SDS or a combination of BOTH the Studies.
- The PC stores the Elemental Time for the work element specified by the Analyst & with a specified PFD allowance, it calculates the STANDARD TIME

Note - • If the SDS used any mathematical formulae then, the SYSTEM must be CAPABLE to accept NEW EQ's, ALTER EQ's or even delete EQ's.

(ii) GENERATION OF METHODS DESCRIPTION:

- Any documentation must include a description of the standard method.

" The STANDARD TIME is only VALID for that STANDARD METHOD!"

- There are several possibilities.

- The possibilities include :

  (1) A library of GENERIC work Elements can be readily RETRIVED, EDITED & made to fit a description.

  (2) EASY ENTRY of NEW DESCRIPTIONS to the LIBRARY.

  (3) Interface via a PMTS / SDS MODUE.

(ii) APPLICATION SPEED RATIO :

- It is a PRODUCTIVITY measure.
- Comparing to the Traditional APPLICATION speed RATIO, the COMPUTERIZED WORK MEASUREMENT technique must have LOWER APPLICATION Speed RATIO.
- The Lower Application Speed Ratio is used to JUSTIFY the Initial Investment & Operating COSTS.
- The computerized SDS / PMTS is 3x or 5x faster than the Traditional SDS/PMTS.

Q. What are the Advantages that lead to a lower value of Application Speed Ratio?

Ans: The advantages are:

1. AUTOMATIC COMPUTATIONS
2. FAST RETRIVAL of DATA
3. FEATURES of USER FRIENDLY SOFTWARES.

⇒ (b) STANDARDS DATABASE FUNCTIONS -

The 2nd Principle fn of computerized WM System is to "SERVE as a DATABASE."

Q. What are the features & issues associated with the STANDARD DATABASE function?

Ans- (i) STORAGE & RETRIEVAL CAPABILITIES:

- The database containing existing Standards must be designed as a CENTRALIZED DATA SYSTEM.
- The data must be uniform across plants & it should easy to retrieve the req. data values.

(ii) MASS UPDATING CAPABILITIES :

- The Change in task may lead to UPDATING the STANDARD DATA.
- Computerized System should posses MASS updating capabilities.

(iii) DATA EXPORT FEATURE :

- Other components such as - (i) Product Cost Estimating
  (ii) CAPP
  (iii) MRP
  (iv) MRPII
  (v) ERP etc. may require the time data for their respective functions.
- The Standard Database should include mechanisms for Sharing Data & other modules.

(iv) STANDARDS MAINTENANCE :

- Here it is done to ensure the Standards are PERIODICALY REVIEWED & updated.

⇒ (4) Work Measurement Based on Expert Systems -

Eg : (a) Auto MOST ⇒ H·B·Maynard & Company.

(b) Quick-LINK ⇒ MTM Association.

Q. What is an Expert System?

Ans: It is a computer program that is capable of solving complex problem that would require a human to have years of Education & Experience.

Q. What are the INGREDIENTS required in a computerized Work Measurement System?

Ans: The 3 Ingredients are :

(1) Knowledge Base of an Expert Analyst

(2) Computer-Compatible description of the Task

(3) Capability to apply the Knowledge Base to a given task description.

Q. What is Knowledge Base?

Ans: Knowledge Base contains:

(a) The TECHNICAL Knowledge about the Task

(b) The LOGIC used by successful Analysts.

- The Knowledge & Logic constitute a set of Rules used by the Analyst.
- These rules must be coded into a computer.

Q. What is a Computer - Compatible description?

Ans: It is a description that must contain all the information req. to determine the Standard Time.

This information includes -

1. Sequence of Work Elements
2. Distance moved
3. Work Variable
4. Tooling used etc.

Q. What is the OBJECTIVE of Computerized Expert Systems?

Ans: OBJECTIVE = "TO ELIMINATE the NEED for the HUMAN EXPERT."

Q. What is meant by "The capability of applying the knowledge base to the given description"?

Ans: It means that the Expert System uses its Knowledge & Logic to solve to solve a specific work measurement problem.

"This problem solving procedure is called "INFERENCE ENGINE." in the Terminologies of Expert Systems.

Q. What are the ADVANTAGE of Expert Systems?

Ans: The advantages include -

1. Ability to transfer the Expert's Knowledge to the staff.
2. Reduces Average costs for Standard Development.
3. INCREASE the Application Speed & further reduces costs.
4. Human Expert's time is now devoted to other imp. fns.

⇒ (5) Maintenance of Time Standards -

- The actual method may be either "LESS" productive or "MORE" productive.

↳ When Actual Method ⇒ Less Productive :

- Here the worker is unable to match the set 100% Efficiency.
- This is usually due to LACK of training.
- Corrective Action : To provide proper instructions & Training about the Standard Method.

↳ When Actual Method ⇒ More Productive :

- Here the req time is far greater than the actual task time.
- This case is difficult to identify as the worker tries to conceal it.
- Here the INGENUITY & CREATIVITY of the worker must not be underestimated.

PTO.

Q. What could the possible improvements be?

Ans: The improvements could include -

1. Improvisation a special tool.
2. Workplace layout is revised
3. External Elements have been made to Internal Elements.
4. Usage of Hand is more.
5. Settings of the m/c is changed to increase the machining time.

Q. What are the other official & legitimate changes?

Ans: The other official & legitimate changes are -

1. Engineering changes in the product
2. Installation of faster Production Machine
3. Improved Work Holding Fixtures.

- All these changes are favourable since they increase productivity reduce costs & increase competativeness.

Q. What is Creeping Changes?

Ans: It is defined "The improvements which are small & have negligible effect on time."

- Even though each change may be insignificant A SUCCESSION of Creeping Change will lead to a SIGNIFICANT change.

Q. What is the purpose of the Standard Maintenance Program?

Ans: PURPOSE = "TO PROTECT the Validity of time standards by Periodically reviewing the tasks.

Q. How to Implement a Standard Maintenance Program?

Ans: • It consists of conducting PERIODIC AUDITS.

- These audits consists of observing the present & Standard methods.
- The comparision is done by WORK ELEMENT-by-WORK ELEMENT

Q. What must be reviewed in an Audit?

Ans: A Through Audit reviews -

1. Machine Speeds
2. Changes in Equipment
3. Engineering Changes
4. Frequency of Irregular Elements

Q. What is the 5% Rule?

Ans: It is a Rule which means "That unless the Time Standard is affected by more than 5%, it remains the same".

- FREQUENCY of AUDIT = Relative IMP of the Task.
- More IMPORTANT the Task = More FREQUENTLY it is AUDITED

Q. What is Standards Erosion & what are its problems?

Ans: Standards Erosion is a consequence of NOT having a MAINTENANCE program. The problems occuring due to Standard's Erosion include -

(a) Reduced Labour Productivity

(b) Labour Cost Rises.

© Losses in Competative Edges

(d) Difficult to Estimate workloads

Q. What is the Table for recommended Frequency for Auditing ? Who introduced it ?

Ans : Niebel introduced the Recommended frequency for Auditing Standards.

The table is : -

TABLE 1 Recommended Frequency for Auditing Standards

Standard Hours per Year (Standard Time × Number of Pieces Processed)	Frequency of Audit
More than 700 hr	6 months
Less than 700 hr but more than 100 hr	1 year
Less than 100 hr but more than 50 hr	2 years
Less than 50 hr	3 years

Source: [2].

(II) LEAN PRODUCTION

⇒ Introduction : It was a Post War strategy by Japan

- These approaches are aimed at :
  1. Increasing Productivity
  2. Reducing Cost
  3. Improve Quality
- These objectives are the same even in methods engineering.
- It represents a MORE Contemporary view of work
- These new approaches can be represented in 2 ways :
  1. Lean Production : It is a Highly Efficient Production System developed by TOYOTA PRODUCTION SYSTEMS (TPS)
  2. Six Sigma : It is Quality Improvement procedure that seeks perfection.
- One Distinctive Characteristics here is that the active participation of the worker must be Enlisted

Lean Production : It means "DOING MORE WORK WITH FEWER RESOURCES"

- It is a Mass Production step up for smaller spaces, lesser work times, fewer workers, & less Equipment
- Lean production : Means giving customers what they want & satisfying or surpassing their expectations.

⇒ History of Lean Production -

- In TPS, "TAIICHI OHNO" was credited ☆ developing Lean Production in TPS.
- Ohno knew that TOYOTA's plant would need to be more flexible. & the space was very limited.

⇒ MUDA = Waste
(Japanese)

- Then he & his colleges proceeded to perfect the basic ideas of Lean Production & which developed -
  ① JIT ② KANBAN System.
  ③ Production Levelling ④ Quality Circles.
- "THE MACHINE THAT CHANGED the WORLD" here the term "Lean Production" was in the SUBTITLE.

- Ohno did not himself coin the term "Lean Prod"
- He titled his book as "The TPS : Beyond Large - Scale Production".
- The Term "Lean Production" was coined by the Researchers @ MIT, to explain the success of TPS.

⇒ Structure of Lean Production System -

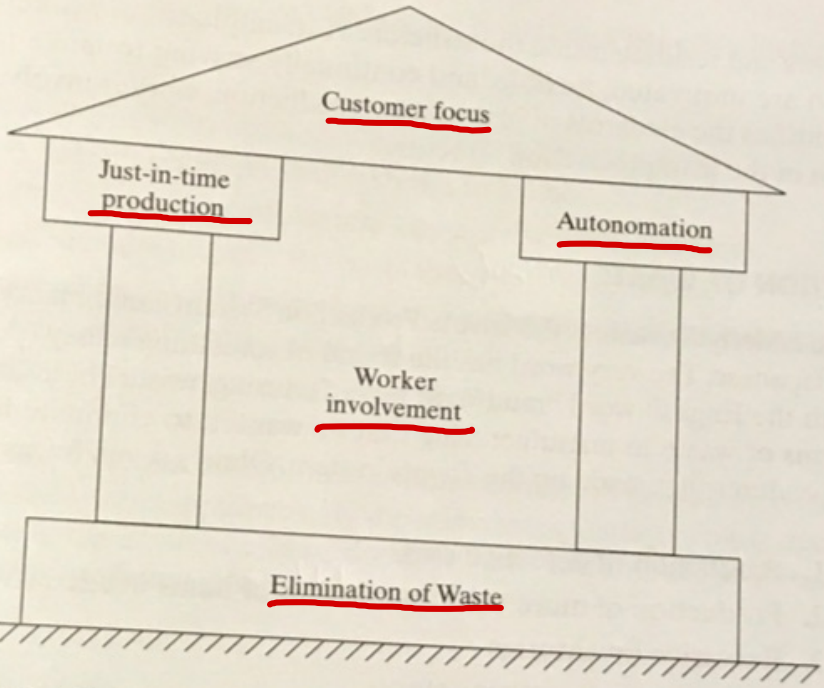


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Figure 1 The structure of a lean production system.

TABLE 1 The Elements of Just-in-Time Production, Worker Involvement, and Autonomation in the Lean Production Structure

MOST IMP

Just-in-Time Production	Worker Involvement	Autonomation
Pull system of production control using kanbans	Continuous improvement (kaizen)	Stop the process when something goes wrong (e.g., production of defects)
Setup time reduction for smaller batch sizes	Quality circles	Prevention of overproduction
Production leveling	Visual management	Error prevention and mistake proofing
On-time deliveries	The 5S system	Total productive maintenance for reliable equipment
Zero defects	Standardized work procedures	
Flexible workers	Participation in total productive maintenance by workers	

Q. What are the Ingredients of a Lean Production System?

Sol: The Ingredients are -

① Base = ELIMINATION of WASTE

② LEFT Pillar = JIT Production.

③ Right Pillar = AUTONOMATION

④ Roof = CUSTOMER FOCUS
It symbolizes focus of the Customer.

⑤ Inside the Structure = WORKER INVOLVEMENT.

- Beneficiary = CUSTOMERS.

⇒ OBJECTIVE of Lean Production is -

" Customer Satisfaction "

- AUTONOMATION = AUTOMATION ≅ a human touch.

⇒ ① Base = ELIMINATION of WASTE -

Here

MUDA = Elimination of waste.

Here MUDA in Japanese is closely related to MUD in English.

↳ The 7 types of waste identified by Ohno are -

① Production of DEFECTED PARTS.

② Production of ITEMS MORE THAN NEEDED. ie OVERPRODUCTION.

③ EXCESSIVE INVENTORIES

④ Unnecessary PROCESSING Steps.

⑤ Unnecessary MOVEMENT of PEOPLE.

⑥ Unnecessary TRANSPORT & HANDELING of MATERIALS.

⑦ WORKER'S WAITING.

↳ ① Production of Defective Parts - (Waste = 1)

- It requires a Quality Control system.
- In mass production QC is typically defined in terms of an ACCEPTABLE QUALITY LEVEL (AQL)

- ie That a minial fractions of defects are Satisfactory
- But for Lean Production Perfect Quality is Req.
- The JIT disciplin requires a "ZERO DEFECT LEVEL" in the quality.

∵ A defective part pushed DOWNSTREAM, means Production would STOP.

- In a Lean System ⇒ There is LITTLE to NO INVENTORY to act as a BUFFER.
- In Mass Production these Buffers are used if these quality problems occur.
- Hence in Lean System, Workers focus on their own units so as to ensure that defective parts are NOT pushed downstream.

↳ (2) Overproduction & Excessive Inventories.
(Wastes = 2 & 3)

Q. What is the Greatest form of Waste according to Ohno?

Ans: According to Ohno "EXCESSIVE INVENTORIES" are the Greatest form of Waste.

- There is a Increase in cost of :

  1. Warehouses
  2. Additional Workers
  3. Storage Equipment
  4. Interest Payments for the above due to Excessive Production & Inventories.

Q. What was done to avoid Excessive Production & Inventories ?

Ans: The KANBAN System provides a CONTROL Mechanism to produce only the Minimum quantity of Parts req for the NEXT seq.

↳ (3) Other forms of Waste :

- Wastes = 4, 5, 6, 7 ⇒ INEFFICIENT USE OF RESOURCES.

- Most of the ineffeciencis can be corrected by Good Methods Engg & Facility Layout.
- Good Equipment Maintance reduces Waiting Time of Workers.

↳ (4) UNNECESSARY PROCESSING STEPS :

- It means that Energy is being expended by worker/machine to finish a product that adds NO VALUE to the product.

↳ (5) UNNECESSARY MOVEMENT of WORKER & MATERIAL :

- It means the movement of worker & materials adds No value to the end product.

Q. Why do people & material move unnessarily ?

Sol: The following reasons are -

1. Inefficient Workplace Layout -
2. Inefficient Plant Layout -
3. Improper Material Handeling Method -
4. Improper Spacing of Production Machines
5. Larger Equipment than req. for a Task.
6. Conventional Batch Production.

↳ (6) WORKER'S WAITING -

- It is the 7th form of MUDA.
- It means NO WORK (either value adding or non-value adding work can be performed).

Q. When are workers forced to wait?

sol: ① Unavailability of Material

② When Assembly Line Has Stopped.

③ Machine is to be Repaired

④ Machine is performing its AUTOMATIC cycle.

⇒ ADDITIONAL INFO:

① Downstream Activity: Toward CUSTOMER

② Upstream Activities: Towards PRODUCTION UNIT.

③ PUSH systems: It is a system that produces Goods based on MARKET demand.

④ PULL systems: It is a Lean Manufacturing Strategy used to reduce waste in production.

⑤ Production Levelling = HEIJUNKA: Used to reduces MURA & in turn reduce MUDA

⑥ Zero Defect: "Think the next operator = CUSTOMER then there is Zero Defect"

⇒ ② Left Pillar : JUST IN TIME Production :

- JIT systems were developed to MINIMIZE inventories, espically in WORK-IN-PROCESS (WIP)
- Excessive WIP was a waste in TPS which was to be Eliminated.
- The IDEAL JIT system produces & delivers exactly the req. no. of each component to the DOWNSTREAM operations.
- It minimizes : ① WIP
  ② Lead Time
  ③ Space & Money
- In TPS, JIT was applied to both production & supplier delivery operations.
- In America JIT is k/s "CONTINOUS FLOW MANUFACTURING".
- Here "CONTINOUS FLOW MANUFACTURING is a form of JIT but & BATCH SIZE = 1 unit".
- Before JIT, there was JUST-IN-CASE. that is to hold large in process inventories to cope with production problems.
- Principal OBJECTIVE of JIT "REDUCE INVENTORIES".

↳ 3 Requisits for Successfuly JIT systems -

Q. What are the requisites to have a Successful JIT system ?

Sol: The 3 requisites are :

① Pull System of Production Control

② Setup Time Reduction for Smaller batches.

③ Stable & Reliable Production Operations.

↳ ① : PULL SYSTEM of Production Control :

- JIT is based on a PULL SYSTEM. ie when the order to make & delivers the parts comes from DOWNSTREAMing stations.
- When the supply starts to get depleted a order is sent to the UPSTREAM station to restock.
- Only when this order is sent, then the UPSTREAM is authorized to do the needful.
- TPS implemented its Pull system o the use of KANBANS.

KANBAN = Card

q. Schematically what is the difference b/w Push & Pull Production Controls ?

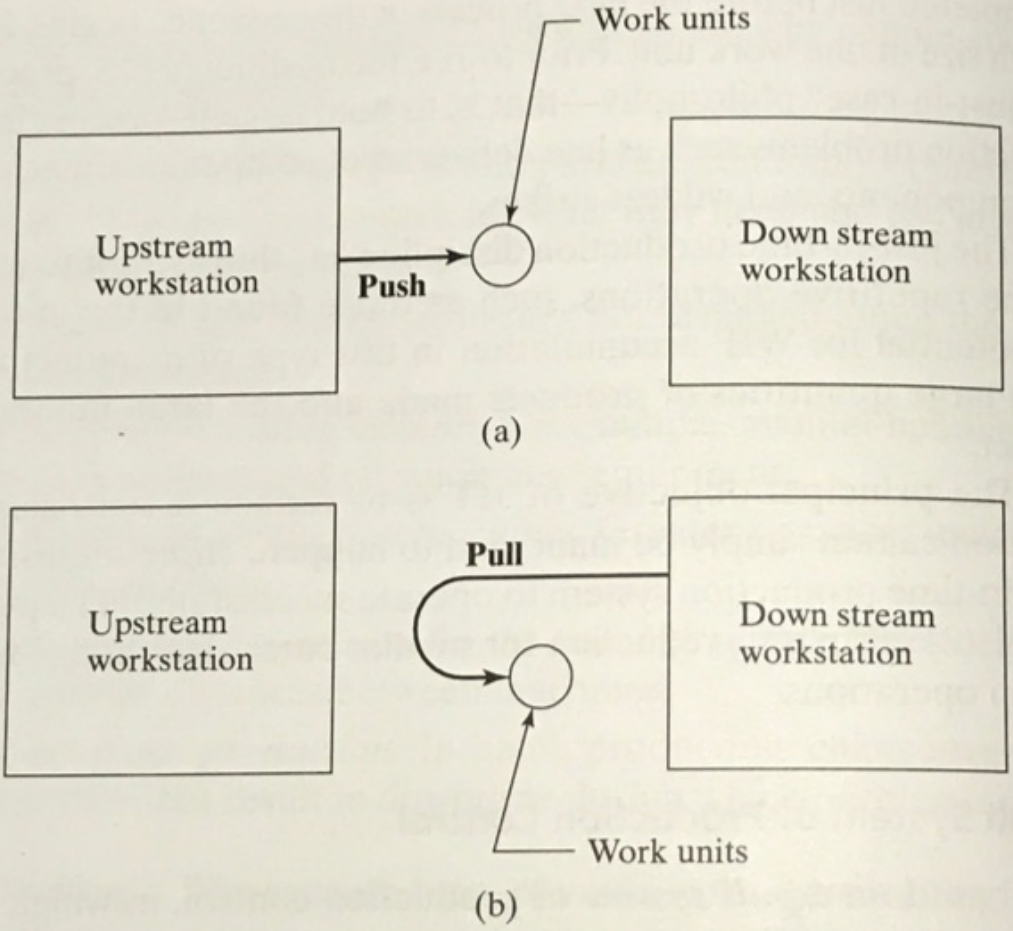

Figure 2 The difference between push and pull systems of production control: (a) push system, in which upstream stations push work to downstream stations and (b) pull system, in which downstream stations pull work from upstream stations.

- There are 2 types of KANBANS:

① Production KANBAN ie (P-KANBAN)
- It authorizes the UPSTEAM to produce the batch of parts.

② Transport KANBAN ie (T-KANBAN)
- It Authorizes transport of the containes of parts to the downstream station.

↳ Explanation of the Kanbans -

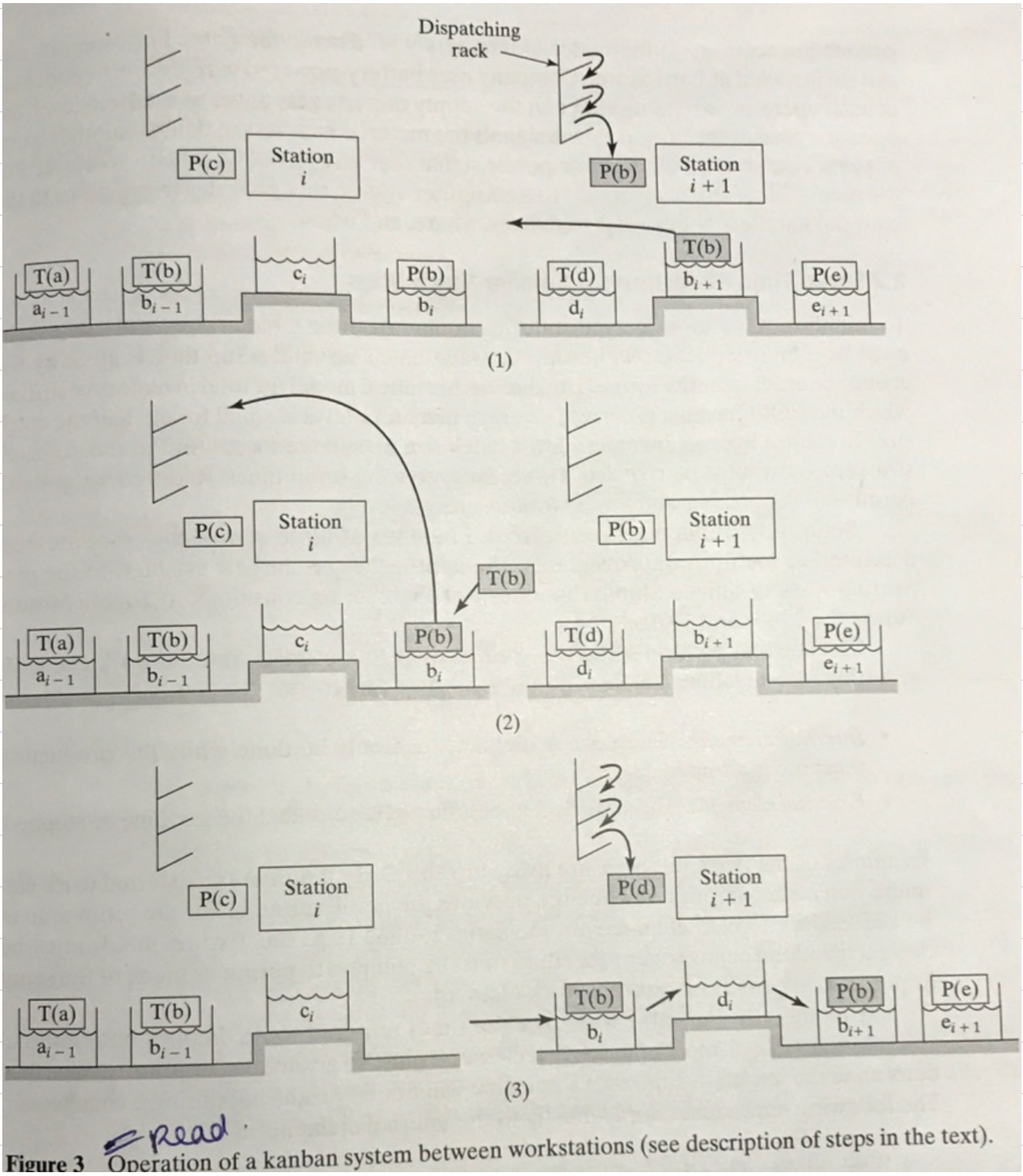

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Figure 3 Operation of a kanban system between workstations (see description of steps in the text).

- How i = UPSTREAM STATION & i+1 = DOWNSTREAM STATION

- Now-a-days KANBANS rely more on Modern communication technologies rather than cards.
- Eg - KANBAN systems at FORD motor company.

↳ ② SETUP TIME REDUCTION for Smaller Batch Sizes:

- To reduces the WIP inventories, BATCH SIZE & SETUP Times must be minimized.
- It was Developed by "SHIEGO SHINGO".
- The relation b/w Setup time, batch size is given by EOQ ie (Economic Order Quantity).
- In EOQ, Average Inventory Level = $\frac{1}{2}$ x Batch Size
- Reduced Setup Times => Smaller Batches => Lower WIP levels.

Q. How to reduce the Setup Time?

Sol: ① IDENTIFY the WORK ELEMENT.
They are of 2 types -

(i) Internal WE : WE can only be done when M/C is STOPPED.

(ii) External WE : WE is done SIMULTAN-OUSLY.

Egs are -

TABLE 2 Examples of Internal and External Work Elements During a Production Setup or Changeover

Internal Work Elements	External Work Elements
Removing tooling (e.g., dies, molds, fixtures) used in previous production job from machine	Retrieving tooling for next job from tool storage
Positioning and attaching tooling for next job in machine	Assembling tooling components next to machine (if tooling consists of separate pieces that must be assembled)
Making final adjustments and alignments of tooling	Reading engineering drawings regarding new setup
Trying out the setup and making trial parts	Reprogramming machine for next job (e.g., downloading part program for new part)

- Here Internal WE must be given HIGHER PRIORITY since it determines the length of Time.

Q. What are few steps to reduce the Internal WE?

Sol: ① Use Time & Methods Study to reduce the sum of Internal WE.

② To use more workers in parallel to perform the setup

③ Eliminate Adjustments

④ Use Quick Acting Fasterns

⑤ Use U-Shaped workers

- The Economic Impact of the Setup Time Reduction can be Evaluated using the Economic Order Quantity Equations.
- The OBJECTIVE of Setup Time Reductions is:

"SINGLE MINUTE EXCHANGE."

↳ ③ STABLE & RELIABLE PRODUCTION Operations -

- The other requirements for a successful JIT include -

Q. What are the other requirements to have a successful JIT, excluding Pull systems of Production Control, Setup Time Reduction for smaller batches & stable & reliable production operation

Sol: The other requirements are -

① PRODUCTION LEVELLING

② ON TIME DELIVERY

③ DEFECT - FREE COMPONENTS

④ RELIABLE PRODUCTION EQUIPMENT

⑤ Capable, Committed, Co-operative WORKFORCE

⑥ DEPENDABLE SUPPLIER BASE

↳ ⓐ Production Levelling :

- PERTURBATIONS in downstream operations tends to be magnified in Upstreams operations.
- Maintaing a "CONSTANT MASTER PRODUCTION" schedule, leads to disturbances in production to be minimized.
- The TROUBLE = Demand of Final Product is NOT CONSTANT.

- Hence the Production System make adjust to the market-place using "PRODUCTION LEVELLING".

Q. What do you mean by PRODUCTION LEVELLING?

Sol: It means distributing the changes in product mix & quantity as evenly as possible over time.

Q. How can Production Levelling be achieved?

Sol: The following approaches may be used -

(1) Authorizing OVERTIME during BUSY periods

(2) Using finished product Inventories to absorbs the fluctations in demand.

(3) Adjusting the cycle time of the Production operation.

(4) Producing in Smaller batch sizes.

- IDEALLY Batch SIZE is reduced to 1.

Q. What are the benefits of Production Levelling?

Sol: Benefits are - (1) Shorter Lead Times

(2) Smaller in process & finished goods inventory.

↳ (b) On-Time Deliveries, Zero Defects, & Reliable Production Equipment.

- For smaller lot sizes used in JIT, parts must be received before stock outs occur at the downstream stations
- JIT requires HIGH QUALITY in every aspect of production.
- Low WIP leaves little room for equipment stoppage.

↳ (c) WORKER'S FORCE & SUPPLIER BASE.

- Workers in JIT must be COMMITTED. CROSS TRAIN & co-operative.
- Lesser batch sizes means that WORKER's must be willing to perform a variety of tasks.

Q. What are the Policies to deal with the vendors for JIT?

Sol:- The new policies are -

① Reduce the Total no. of Suppliers

② Enter into Long-Term Agreements.

③ selecting parts suppliers which are Nearby.

⇒ ③ Right Pillar = AUTONOMATION :

Defined by Ohno as "AUTOMATION & a HUMAN TOUCH."

- The aspect of AUTONOMATION is that the M/C's & processes are designed to be mistake-proof.

Q. What are the ASPECTS of AUTONOMATION ?

Sol- The 3 aspects of AUTONOMATION are -

① Stopping the Process

② Error Prevention ie "POKAYOKE"

③ Total Productive Maintainence.

↳ ① STOPPING the PROCESS :

JIDOKA = They are M/C's which STOP automatically when something goes WRONG.

- In TPS, production machines were equipped & automatic stop devices that activate when a defected unit is produced.
- Adjustments must be made to fix the machine & thus improving overall quality.

IMP • Apart from the QUALITY CONTROL fn, AUTONOMATION also refers to the M/c which are controlled to STOP production when the required QUANTITY has been completed.

Thus preventing OVER PRODUCTION.

Q. What are the various devices used in AUTONO-MATION?

Sol: It consists of the following Control device -

1. A SENSOR - To detect Abnormal operation that could result in a defect.
2. A device to COUNT the NO. OF parts PRODUCED.
3. A medium to STOP the M/c or production line.

- In plants where AUTONOMATION does not exist, a machine requires CONTINOUS HUMAN ATTENDENCE to monitor its operations.
- However AUTONOMATED machines do not req. continious HUMAN attendance & hence 1 worker can oversee the operation of multiple machines. & thus improve Worker productivity.

- Workers must be willing & be able to develop a greater variety of skill.
- The net effect of more versatile workers is that the plant becomes more FLEXIBLE in its ability to shift workers amongst machines during a WORKLOAD MIX.
- AT TPS, JIDOKA concept is Extended to its "FINAL ASSEMBLY LINES".
- Downtimes = Idle Time = on a Final Assembly line may be EXPENSIVE.

↳ (2) ERROR PREVENTION:

- This concept is derived from 2 Japanese words-

(i) POKA = ERROR

(ii) YOKE = PREVENTION

∴ POKA YOKE = ERROR PREVENTION

- Concept of POKA YOKE was developed by "SHIEGO SHINGO".
- Use of POKAYOKE devices relieve the worker of constantly monitoring the process for errors.

Q. What are few examples of production defects due to mistakes in manufacturing.

Sol: Egs are -

1. OMISSION of PROCESSING STEPS.
2. Incorrectly locating a WORK PART
3. Using the wrong tool.
4. NOT aligning JIGS
5. Neglecting to Add a Component.

Q. How does Poka-yoke respond to the errors that have been identified?

Sol- It responds in either of the 2 or both ways:

1. STOPS the PROCESS: The automated cycle is stopped when a problem is detected in the operation of a production M/c.
2. PROVIDES an ALERT: It provides an AUDIBLE or VISIBLE Error Sign that an error has occured.

"ANDON BOARDS is a means of implementing this type of Response"

↳ ③ TOTAL PRODUCTIVE MAINTENANCE :

- The production equipment in TPS is Highly Reliable.
- JIT systems CANNOT tolerate machine breakdowns.
- Lean Production requires an Equipment maintenance program that MINIMIZES machine breakdowns.
- TPM = Total Productive Maintenance is a CO-ORDINATED group of activities whose objectives are :

OBJECTIVE'S are : MINIMIZE PRODUCTION LOSSES due to EQUIPMENT failure, & malfunctions.

Q. What are the different types of Maintenances ?

TABLE 4 Some Maintenance Definitions

	Term	Definition
①	Emergency maintenance	Repairing equipment that has broken down and returning it to operating condition. Action must be taken immediately to correct the malfunction. Also known as reactive maintenance.
②	Preventive maintenance	Performing routine repairs on equipment (e.g., replacement of key components) to prevent and avoid breakdowns.
③	Predictive maintenance	Anticipating equipment malfunctions before they occur based on computerized machine monitoring, machine operator being attentive to the way the machine is running, historical data, and other predictive techniques.
④	Total productive maintenance	Integration of preventive and predictive maintenance to avoid emergency maintenance.

Q. What is meant by AVAILABILITY ?

sol: The Traditional measure of machine reliability is it's "Availability".

- It is the proportion of the Total Desired operating time, that the machine is ACTUALLY AVAILABLE & OPERATING

- To be more PRECISE:

"AVAILABILITY is defined in terms of 2 other Reliability measure, ie:

① MEAN TIME B/W FAILURES (MTBF)

② MEAN TIME TO REPAIR (MTTR)

Q. What is meant by MTBF & MTTR ?

sol: ① MTBF : It indicates the average length of time the piece of Equipment runs b/w Breakdowns.

② MTTR : It indicates the Average Time Required to service the Equipment & put it back into operation.

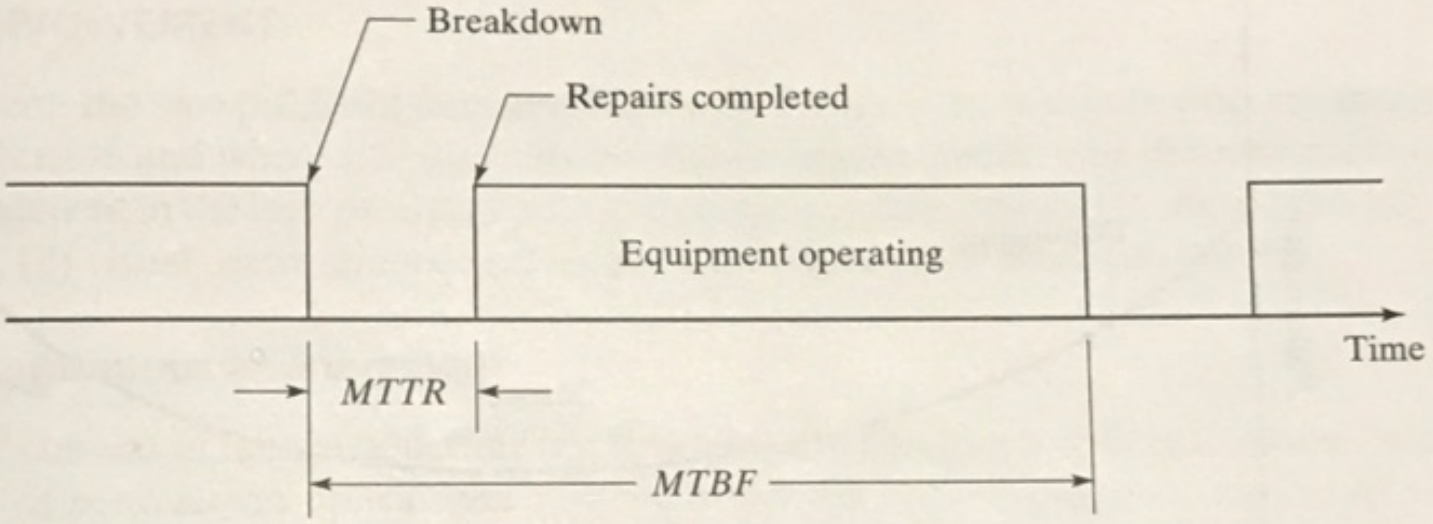

Figure 4 Time scale showing MTBF and MTTR for a piece of machine that breaks down occasionally.

∴ AVAILABILITY, $A = \frac{MTBF - MTTR}{MTBF}$

- Availability is Typically Expressed as a Percentage.

Q. What are the other reasons besides Breakdown that lead to operating the Equipment at a level less than its Full Capability?

Sol: The others may include:

1. Low Equipment Utilization
2. Production of Defective Parts
3. Operating at a lesser speed than designated.

- As a machine ages, its Availability reduces.
- Hence a "U-Shaped curve for AVAILABILITY as a fn of Time over the life of the Equipment" is seen.

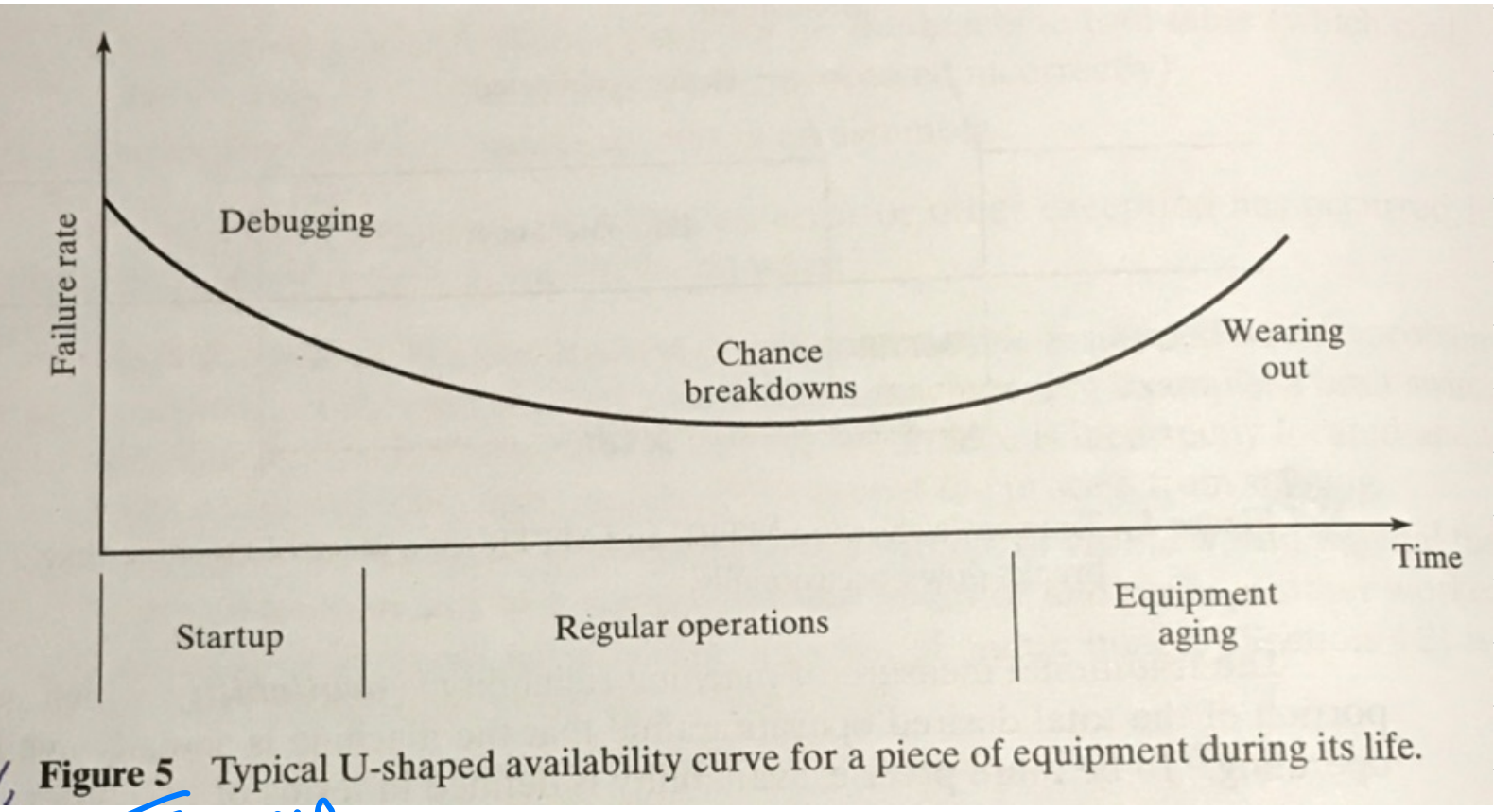

Figure 5 Typical U-shaped availability curve for a piece of equipment during its life.

IMP.

Q. What is Utilization?

sol: UTILIZATION: Refers to the amount of output of a machine during a given time period relative to its CAPACITY during that same period.

ie $$U = \frac{Q}{PC}$$

where U = Utilization
Q = Actual quantity produced
PC = Production Capacity.

Def: Utilization can also be defined as "No. of hours of Productive Operations relative to the Total no. of hours the machine is available."

Q. What could be the reasons for POOR M/c Utilization?

Sol: It includes the following reasons:

1. Poor scheduling of work
2. Machine Starved for work
3. Downtime due to setups
4. Worker Absenteeism
5. Low Demand.

Q. What are other reasons for production of defected parts?

Sol- The other reasons include -

1. Incorrect Machine Settings
2. Inaccurate Adjustments
3. Improper Tooling.
4. Defective Starting Material
5. Human Error.

Q. What is FRACTION DEFECT RATE ?

Sol: FRACTION Defect Rate : It is the PROPORTION of defected parts produced in a given process.

∴ YIELD of a process is given as -

$$Y = 1 - q$$

where Y = Process Yield

ie Process Yield : It is the ratio of conforming parts to the total parts processed.

Q. What is Operating Capabilities ?

Sol: It is the ratio of ACTUAL OPERATING speed to the designed speed of the machine.

ie Operating Capability = π_{os}.

Hence,

$$OEE = A \times U \times Y \times \pi_{os}$$

where OEE = Overall Equipment Effectiveness

A = Availability

U = Utilization

Y = Process Yield

π_{os} = Operating Capability

⇒ Hence OBJECTIVE of TPM is:

" To ensure OEE is as close as possible to Unity (100%)".

⇒ (4) Inside: WORKER INVOLVEMENT:

- It refers to Workers who are MOTIVATED, FLEXIBLE & those who participate in CONTINOUS IMPROVEMENT.
- It consists of 3 Main Topics / Categories / aspects- ie:

(1) CONTINOUS IMPROVEMENT ie (KAIZEN).

(2) VISUAL MANAGEMENT & 5S's.

(3) STANDARDIZED WORK PROCEDURES.

⇒ (1) CONTINOUS IMPROVEMENT

Here KAIZEN = Continous Improvement.

- KAIZEN is implemented by means of worker teams Ms "QUALITY CIRCLES"
- Quality Circles are Organised to address specific problems that have been identified.

- Quality Circles => KAIZEN CIRCLE:

- KAIZEN is a PROCESS that attempts to involve All workers as well as their supervisors.

- Workers can be a member of more than 1 KAIZEN circles.

- OBJECTIVE of KAIZEN circles:
  "Is to solve Problems in production".

- It can also be used to encourage a workers sense of RESPONSIBILITY, Gain ACCEPTANCE & RECOGNITION among colleagues.

- KAIZEN is applied on a Problem-by-Problem basis by a worker team.

- Team members are selected According to their knowledge & Expertise.

- They serve part time to a project team, ie in ADDITION to fulfilling their regular duties.

- They meet for about 2-4 times a month, & each meeting lasting for an hour.

⇒ (2) VISUAL MANANAGEMENT & 5S.

Principle of VISUAL MGT:

"STATUS of the WORK SITUATION should be EVIDENT just by looking at it."

- VISUAL MGT = VISUAL WORKPLACE.
- Objects that Obstruct the view inside the plant are not allowed making the Entire space visible.
- The build up of WIP is limited to a specific height.
- Hence it encourages - (i) Kaizen
  (ii) Good House Keeping.
- Eg of VISUAL MANAGEMENT are KANBAN system.
- Implementing the VISUAL MANAGEMENT involves the use of "ANDON BOARDS".

ANDON BOARD: It is a light panel installed at or above a workstation to indicate the operation status.

- Different colour lights are used to indicate the status of operation.

eg: Green = Normal operation.

Yellow = Worker has a problem & calling for Help.

Red = Line Stoppage.

- VISUAL workplace principle is also used in Worker Training.
- It includes the use of Photographs, drawings, & diagrams.

ie " A picture is worth a thousand words".

Q. What is meant by 5s ?

sol: It is a means of including the workers in the VISUAL WORKPLACE.

It is a SET of Procedures used to ORGANIZE work areas in a plant.

Q. What are the 5s?

sol- (1) SEIRI = SORT

(2) SEITON = Set in Order

(3) SEISO = Shine

(4) SEIKETSU = STANDARIZE

(5) SHITSUKE = SUSTAIN

TABLE 5 The Five Steps in the 5S System

VVVIMP

1. ***Sort.*** Sort things in the workplace; in particular, identify the items that are not used and dispose of them. This eliminates the clutter that usually accumulates in a workplace after many years.
2. ***Set in order.*** Organize the items remaining in the work area after sorting according to frequency of use, and provide easy access to those items that are most often needed.
3. ***Shine.*** Clean the work area and inspect it to make sure that everything is in its proper place. The alternative S-words for this step are "sweep" and "scrub," which may also be required in the work area.
4. ***Standardize.*** Document the standard locations for items in the workplace. For example, use a "shadow board" for hand tools, in which the outline of the tool is painted on the board to indicate where it belongs. Looking at the shadow board, workers can immediately tell whether a tool is present and where to return it.
5. ***Sustain.*** Establish a plan for sustaining the gains made in the previous four steps, and assign individual responsibilities to team members for maintaining a clean and orderly work environment. Make workers responsible for taking care of the equipment they operate, which includes cleaning and performing minor maintenance tasks.

- The 5s is a Technique of VISUAL MGT, which should be followed in SEQUENCE.
- 5s provide an ADDITIONAL means of implementing VISUAL MGT to provide CLEAN, ORDERLY & a VISIBLE Work Environment that promotes HIGH MORALE among workers.
- The 5s system must be a continous process to sustain the accomplishments that have been made.

⟹ (3) STANDARIZED WORK PROCEDURES:

- Std Work procedures were developed at TPS, using approaches similar to traditional methods engg. techniques.

OBJECTIVES of STANDARIZED WORK PROCEDURE-

1. MAINTAIN HIGH PRODUCTIVITY
2. BALANCE the WORKLOAD
3. MINIMIZE Amount of WIP.

- At TPS, a Standarized Work Procedure has 3 Components ie:
    1. CYCLE TIME
    2. WORK SEQUENCE
    3. STANDARD WIP QUANTITY.

- These Components are documented using forms.

⇒ CYCLE TIME & TAKT TIME:

1. CYCLE TIME: It is the Actual Time it takes to complete a TASK.
    - It is measured by a STOPWATCH
2. TAKT TIME: It is the RECIPROCAL of the DEMAND RATE for a product, adjusted for the Available shift time in the factory

∴ TAKT → German Word = CADENCE (or) PACE

ie $$T_{takt} = \frac{EOT}{Qdd}$$

where T_{takt} = Takt Time

EOT = Effective DAILY Operating Time

Qdd = Daily quantity of units demanded.

Here the EOT = Shift hours worked each day without subtracting the allowances

Here the Qdd = It is the MONTHLY demand of the unit divided to the no. of working days in that month.

ie $$Qdd = \frac{\text{MONTHLY Demand}}{\text{No. of working days}}$$

Q. Why is EOT not adjusted for Lost Time & Qdd for defective units?

Sol: It is done so that it draws the attention, so that corrective actions will be taken to Eliminate them.

↳ Eg of Takt Time:

Example 2 Takt Time

The monthly demand for a certain part is 10,000 units. There are 22 working days in the month. The plant operates two shifts, each with an effective operating time of 440 min. Determine the takt time for this part.

Solution: With two shifts, the effective daily operating time is 2(440 min) = 880 min. The daily quantity of units demanded is 10,000/22 = 454.5 pc/day.

$$T_{takt} = 880/454.5 = 1.94 \text{ min}$$

- Takt Time provides a specification based on demand.
- In TPS, the work was so designed that the cycle time is synchronized with the Takt Time.
- This was accomplished through:
  1. Work Sequence
  2. Standardizing the WIP quantity.

⇒ WORK SEQUENCE:

Def: It defines the order of work elements / operations performed by a given worker.

① For a Single Worker & Repetitive Cycles:

- It indicates the actions that need to be carried out.

② For a MULTI FUNCTION worker & Many Machines with their OWN semi-automatic cycle -

- It describes the work to be done on each machine & the order in which the machines must be attended too.

- Pictures & Videos are shown to depict the proper use of HAND tool, Safety practices & correct ERGONOMIC posture.

- For the MULTI MACHINE scenario, the work sequence is documented by the means of the "STANDARD OPERATIONS ROUTINE SHEET".

- This form lists the machines to be visited.

- A HORIZONTAL time scale depicts how long each operation should take.

- Solid Line = Worker.

- Dashed Line = Machine

- Squiggly Lines = Worker walking b/w machine

- It is NOT unusual for a DOZEN steps to be req to complete a Part.
- Cycle Time may differ from the Takt Time.
- In TPS. the work is organized in such a way that the cycle time for each step = Takt Time.
- By matching the cycle Time & the Takt Time. the sequences are balanced & the amount of WIP is MINIMIZED.

Q. How can the work to be done be ORGANISED so that cycle Time = Takt Time?

Sol: In TPS. Tasks are integrated into WORK CELLS.

WORK CELL: It is a group of workers & stations that are physically arranged in sequential order so that parts can be produced in small batches.

- These cells are typically U-SHAPED to promote COMRADESHIP among workers & to achieve continous flow of work.

Eg of Work Sequence -

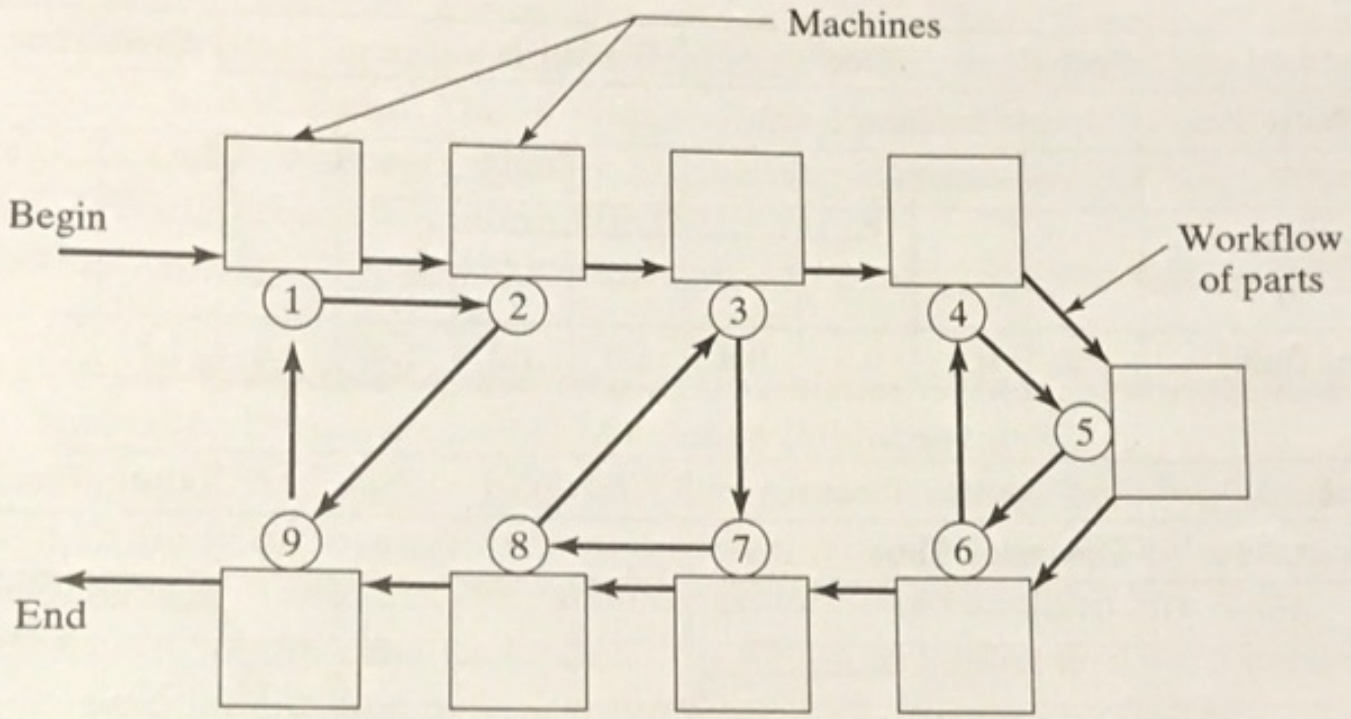

Figure 7 Cell layout with three workers performing nine operations. Operations are indicated by numbers in circles. The first worker performs operations 1, 2, and 9; the second worker performs operations 3, 7, and 8; and the third worker performs operations 4, 5, and 6.

⇒ STANDARD WIP QUANTITY

- WIP is the quantity of work Parts that have STARTED the WORK SEQUENCE but have NOT Completed the WORK SEQUENCE
- WIP consists of both the parts : ie
  1. Parts currently being processed
  2. Parts between processing steps.
- WIP does not account for FINISHED or RAW materials.

- In TPS, the Standard WIP Quantity is the MINIMUM no. of Parts Necessary to AVOID situations where workers are waiting.

↳ Problems on Takt Time :

10 The weekly demand for a certain part is 950 units. The plant operates 5 days/week, with an effective operating time of 440 min/day. Determine the takt time for this part.

Sol: Weekly Demand = 950 units

$\therefore$ Monthly Demand = $950 \times 4 = 3800$ units

No. of working days = $5 \times 4 = 20$ days

EOT = 440 min

$\therefore Q_{dd} = \frac{3800}{20} = 190$ pc/day.

$\therefore T_{Takt} = \frac{EOT}{Q_{dd}} = \frac{440}{190} = 2.31$ mins

11 The monthly usage for a component supplied to an appliance assembly plant is 5000 parts. There are 21 working days in the month and the effective operating time of the plant is 450 min/day. Currently, the defect rate for the component is 2.2%, and the equipment used to produce the part is down for repairs an average of 22 min/day. Determine the takt time for this part.

Sol: Monthly Demand = 5000 parts

No. of days = 21

EOT = 450 − 22 = 428 min/day.

$\therefore Q_{dd} = \frac{5000}{21} = 233 \text{ units/day.}$

$\therefore T_{Takt} = \frac{EOT}{Q_{dd}} = \frac{428}{233} = 1.836 \text{ mins}$

for this part.

12 The monthly demand for a part produced for an automotive final assembly plant is 16,000 units. There are 20 working days in February and the effective operating time of the plant is 900 min/day (two shifts). The fraction defect rate for the component is 0.017, and the automated machine that produces the part has an availability of 96%. Determine the takt time for this part.

Sol: Monthly demand = 16000 units

No. of days = 20 days.

EOT = 864 mins for 2 shifts

$\therefore Q_{dd} = \frac{16000}{20} = 800 \text{ units} - 13.6 = 786.4$

$\therefore T_{Takt} = \frac{EOT}{Q_{dd}} = \frac{864}{786.4} = 1.098 \text{ min}$